

Participation and Spending in the Medicare Shared Savings Program

Job Market Paper

Alberto Cappello

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Abstract

The Medicare Shared Savings Program (MSSP) is a voluntary program that provides incentive payments to healthcare providers that form integrated healthcare organizations, known as Accountable Care Organizations (ACOs). The MSSP rewards ACOs for keeping average per-capita spending below a benchmark based on historical spending by ACO providers under the fee-for-service (FFS) system. I develop and estimate a structural model of ACO participation and performance in the MSSP. I show that blending the ACOs' historical spending with a regional adjustment factor induces adverse selection, enabling some ACOs to earn incentive payments without reducing spending below the FFS level. Moreover, the benchmark rebase-ment, which updates the benchmark over time to reflect observed healthcare spending among ACO's providers, incentivizes ACOs to delay spending reductions to avoid lowering future benchmarks. I propose a revised benchmarking methodology to address these inefficiencies. Counterfactual analyses demonstrate that this alternative policy mitigates adverse selection and significantly increases Medicare savings.

1 Introduction

There is a broad consensus among health economists that the high costs and inefficiencies in the U.S. healthcare system are largely due to the fragmented nature of healthcare delivery and misalignment of incentives between Medicare and healthcare providers (Cebul et al., 2008). Medicare patients are frequently treated by numerous care providers who have only weak organizational ties with one another and often little expertise in coordinating care. This results in duplicate and unnecessary services, heightened error rates, and inadequate care coordination.

Medicare payments operate under a fee-for-service (FFS) system, which lacks incentives for cost-efficient healthcare delivery. Under this model, providers are reimbursed based on the cost and volume of services rendered. As a result, healthcare providers may find it profitable to deliver excessive services or opt for unnecessarily expensive procedures, even when doing so does not necessarily improve patient outcomes (Hendee et al., 2010, Chatterji et al., 2022).

Value-based programs and pay-for-performance contracts are increasingly popular forms of government regulation aimed at correcting the FFS system's adverse incentives and at improving the efficiency of healthcare delivery. The Medicare Shared Savings Program (MSSP) is an incentive payment program administered by the Centers for Medicare and Medicaid Services (CMS) since 2012. The goal of the MSSP is to encourage the formation of integrated provider organizations and make it financially viable for providers to reduce spending through improved care coordination and proper use of preventive care and care monitoring. To participate in the MSSP, a group of independent Medicare providers, including hospitals, physician groups, and individual practices, must form a joint venture referred to as an Accountable Care Organization (ACO). Under the MSSP, ACOs are held accountable for the per-capita expenditure (Medicare Part A and Part B) and quality of care for a defined population of Medicare beneficiaries.

To promote care coordination among ACO providers, the MSSP contract rewards ACOs based on the overall group performance. ACOs qualify for a bonus payment, known as shared savings, if their per-capita FFS spending falls below a predetermined benchmark. This payment is calculated as a fraction of the difference between the benchmark and actual spending. Moreover, ACOs can opt for either a one-sided or two-sided shared savings contract. Under the two-sided contract, ACOs receive a larger share of the savings compared to the one-sided contract, but are also subject to financial penalties if their spending exceeds the benchmark.

The main focus of this paper is the MSSP rules for setting and updating the benchmark and how the latter affect ACOs' participation and spending. When an ACO enters the MMSP, the historical benchmark is calculated as the average per-capita FFS spending of the Medicare beneficiaries

assigned to the ACO. As the ACO progresses in the MSSP, the two adjustments are applied to the historical benchmark. First, the ACO benchmark is recalculated every three years using the average spending of the previous three years (rebased benchmark). Secondly, a regional adjustment factor is applied to the benchmark in proportion to the difference between the ACO rebased benchmark and average regional Medicare spending (regionalized benchmark). For ease of exposition, I will refer to the regionalized rebased benchmark as “benchmark” in this introduction.

To study the impact of the benchmarking rules on the ACOs’ participation and spending, I abstract from the complex interaction between the ACO’s providers and model each ACO as a single agent. Following a standard assumption in the health economics literature, I assume that ACOs’ utility depends on patients’ health benefits and financial profits (Culyer and Newhouse, 2000). Under the MSSP, providers continue to receive payments through the FFS system. ACO’s financial incentives are driven by a combination of FFS reimbursements and shared savings payments. The latter are the results of the ACO’s providers’ efforts to reduce their spending relative to the benchmark. This effort is captured by the quality score, which is an aggregate measure of the level of care coordination, preventive care, and care monitoring. The cost of increasing the quality score varies across ACOs as it depends on the ACO’s characteristics and patients’ health risks. Additionally, many ACOs face substantial entry costs, as they must invest in information technology to enhance care coordination. To estimate the cost of quality score and the entry cost, I develop and estimate a single agent dynamic model. This model assumes that ACOs decide whether to participate in the MSSP or remain under the FFS system and determine the level of quality score required to reduce spending. Finally, I use the estimated model to conduct a series of counterfactual simulations to estimate the effect of benchmark rebasement and regionalization on the overall savings generated by the MSSP.

From Medicare’s perspective, it may be ideal for ACOs to act myopically, that is, to achieve the largest possible spending reduction each year. However, due to the MSSP rebasement mechanism, which recalculates future benchmarks based on an ACO’s previous spending, ACOs may find it advantageous to reduce their efforts (and generate smaller savings) in earlier years to secure more favorable benchmarks in subsequent periods. To reduce the influence of an ACO’s past spending on its benchmarks, Medicare introduced a regional adjustment factor. Starting in 2017, the MSSP began blending the ACO’s rebased benchmark with the FFS spending of Medicare providers in the same service area while adjusting for differences in patient population health risks.

The benchmark regionalization is not an ideal solution because it can result in adverse selection. I present empirical evidence that ACOs with a positive regional adjustment are more likely to select in the MSSP, while ACOs with a negative regional adjustment tend to select out of the MSSP.

Moreover, the counterfactual analysis reveals that some ACOs participate in the MSSP and obtain shared savings payments without reducing their spending relative to the FFS level. Removing the benchmark regionalization reduces Medicare spending by 57\$ per-capita relative to the status quo and approximately 85% of this reduction is the result of lower shared savings payments.

To assess the impact of benchmark rebasement, I simulate a counterfactual scenario where the ACOs' spending contributes to the regional adjustment calculation, but does not directly influence future benchmarks. Under this scenario, reduction in spending is larger in the early years of the participation period and then levels off over time, which is the opposite of what is observed in the actual MSSP. This suggests that benchmark rebasement incentivizes ACOs to delay cost-containment efforts to avoid reducing their future benchmarks, a behavior commonly referred to in the literature as the "Ratchet Effect". According to counterfactual results, removing the benchmark rebasement reduces overall Medicare spending decreases by 117\$ per-capita.

Motivated by the need to better balance the trade-off between rebasement and regionalization, I propose an alternative benchmarking rule called "Conditional Regionalization". To address adverse selection, this rule modifies the current MSSP methodology by imposing that to qualify for a positive regional adjustment, ACOs are required to that have generated savings. Furthermore, to encourage participation of ACOs whose historical spending exceeds the regional average, this rule waives the negative regional adjustment for ACOs that achieve savings.

Counterfactual results indicate that this approach effectively leverages benchmark regionalization to mitigate the ratchet effect while simultaneously addressing the problem of adverse selection. Medicare savings increase by 2.02% relative to the status quo, which amounts to approximately 15.6 billion dollars. Moreover, this additional spending reduction is driven by both existing ACOs and new entrants participating in the MSSP under the alternative policy.

This paper contributes to the vast economics literature on the impact and optimal design of financial incentives for healthcare providers (e.g., [Cutler, 1993](#); [Gaynor et al., 2004](#); [Clemens and Gottlieb, 2014](#); [Ho and Pakes, 2014](#); [Einav et al., 2022](#); [Eliason et al., 2018](#); [Hackmann, 2019](#)). Previous works have analyzed specific Medicare programs with objectives similar to the MSSP. In particular, the Bundled Payments for Care Improvement Initiative (BPCI) is a payment model in which Medicare providers receive a single payment (benchmark) for a collection of services related to a specific episode of care. [Einav et al. \(2022\)](#) show the impact of adverse selection in the context of the BPCI and investigate the effects of alternative benchmarking policies. [Zhang et al. \(2016\)](#) studied the behavior of hospitals under the Hospital Readmissions Reduction Program, a mandatory program that penalizes hospitals that do not reduce readmissions below target levels.

More narrowly, this paper contributes to the literature on the effect of the Medicare Shared

Savings Program. [Reddig \(2023\)](#) estimates a static model of ACOs’ optimal savings and quality choices and shows a substantial trade-off between reducing costs and increasing quality. [Frandsen and Rebitzer \(2015\)](#) calibrate a simple model of ACO performance and find free-riding incentives that significantly reduce ACO providers’ efforts to reduce spending. [Aswani et al. \(2019\)](#) examine the role of asymmetric information between Medicare and ACOs and propose an alternative contract design. They consider both selection on the cost of spending reduction and selection on the historical spending. However, since they only use data for 2015, which is two years before the benchmark regionalization becomes effective, they cannot account for the negative impact that adverse selection will have on Medicare spending when the regionalization takes effect. Furthermore, their single-period model cannot capture the dynamic implications of benchmark rebasement. Hence, we cannot tell how their alternative contract affects participation and the incentives to generate savings over time.

Compared to the existing literature, this paper makes three key contributions. First, it develops a multi-period model to analyze ACOs’ participation and spending decisions over time. Second, it is the first to estimate and evaluate the dual effects of the ratchet effect caused by benchmark rebasement and adverse selection arising from benchmark regionalization. Finally, it proposes an alternative benchmarking rule that can mitigate the ratchet effect and prevent adverse selection, while also substantially improving the savings generated by the MSSP.

The rest of this paper is organized as follows: Section 1 gives an overview of the Medicare Shared Savings Program and describes how benchmark rebasement and regionalization work. Section 2 describes the data sources and presents evidence of adverse selection. In Section 3, I discuss the utility function of an ACO and present a simplified version of the dynamic model of ACO participation and quality choice to formally describe the ratchet effect and how it is affected by benchmark regionalization. In Section 4, I present my estimation strategy and discuss the identification of the model primitives. In Section 5, I show the results of the counterfactual analysis, and Section 6 concludes.

2 Institutional Setting

Proponents of the MSSP argue that providing a financial incentive to reduce spending below the FFS level will improve care coordination among providers and reduce unnecessary healthcare utilization. To participate in the MSSP, healthcare providers form Accountable Care Organizations, joint ventures created to improve care coordination among independent providers. Nearly any Medicare provider can start or participate in an ACO, including individual physicians, group practices, and large hospital systems. Medicare fee-for-service (FFS) beneficiaries are then assigned to ACOs by Medicare according to their primary care provider (PCP). ACOs can enter the Medicare Shared Savings Contract with CMS and earn a bonus payment based on their performance in terms of average per-capita spending and quality score. This section will first present the one-sided and two-sided shared savings contracts. Secondly, I will describe the benchmarking methodology in detail, focusing in particular on benchmark rebasement and regionalization. Finally, I will discuss the quality score and how I will use it in this paper.

2.1 The Medicare Shared Savings Contract

Under MSSP, providers who are members of an ACO are still reimbursed by Medicare according to the standard fee-for-service system. Hence, providers still receive the marginal payment for each service, and therefore, the FFS reimbursement is still proportional to the volume of services. The MSSP gives providers financial motivation to integrate care delivery by creating an incentive payment that depends on the ACO's performance. Each ACO is assigned a per-capita spending benchmark, which should represent the counterfactual Medicare FFS spending on the ACO's assigned beneficiaries without the MSSP. At the end of the year, each ACO can be eligible to earn an additional bonus payment called shared-savings if the average per-capita spending of the ACO's providers is below the spending benchmark. In this subsection, I will describe the shared savings contract using the general term benchmark, and in the following subsection, I will introduce the benchmark rebasement and regionalization.

Let y denote the average per-capita FFS spending of the Medicare beneficiaries assigned to the ACO, and let $q \in [0, 1]$ be the CMS measure of the ACO's effort at reducing spending through higher care quality as measured by the ACO's performance in terms of care coordination, preventative care, and care monitoring. The shared savings payment obtained depends on the spending performance relative to the benchmark b and on the quality score q . The ACO's spending perfor-

mance is evaluated in terms of the saving rate

$$s = \frac{b - y}{b}$$

ACOs earn a fraction $\psi q \in [0, 1]$ of the spending reduction relative to the benchmark $(b - y)$ if the savings rate is above the minimum saving rate $\underline{s}$. Notice that the fraction of savings paid to the ACO depends on the shared savings rate $\psi = \frac{1}{2}$ and also on the quality score $q \in [0, 1]$. This contract is called one-sided since the ACO does not incur any penalty or loss if spending is above the benchmark. The shared savings payments under the one-sided contract are given by the equation [Equation 1](#)

$$SS_1(y, q) = \begin{cases} \frac{1}{2}q(b - y) & \text{if } \frac{b-y}{b} > \underline{s} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The MSSP gives ACOs also the option to choose a two-sided contract that has both upside and downside risk. This contract has higher shared savings rate ($\psi = 0.75$), but requires the ACO to pay a penalty if its spending is larger than the benchmark. The penalty is given by a fraction $(1 - \psi)(1 - q)$ of the excess spending relative to the benchmark if the savings rate is lower than $-\bar{s}$. The shared savings payments under the one-sided contract are given by the equation [Equation 2](#).

$$SS_2(y, q) = \begin{cases} \frac{3}{4}q(b - y) & \text{if } \frac{b-y}{b} > \underline{s} \\ \frac{1}{4}(1 - q)(b - y) & \text{if } \frac{b-y}{b} < -\bar{s} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Panel (a) in [Figure 1](#) shows the one-sided shared savings contract. The one-sided shared savings payments is a non-increasing function of spending. Indeed, the shared savings are zero if spending is above $b(1 - \underline{s})$ and the slope of the shared savings function is $-0.5q$ when the shared savings payments are positive. Panel (b) in [Figure 1](#) shows the two-sided shared savings contract. The two-sided shared savings payments is a non-increasing function of spending. Indeed, the shared savings payments are zero if spending is between $b(1 - \underline{s})$ and $b(1 + \bar{s})$, and the slope of this function is $-0.75q$ if spending is below $b(1 - \underline{s})$ and $-0.25(1 - q)$ if spending is above $b(1 + \bar{s})$. Therefore, the two-sided contract implies stronger incentives to keep spending below the benchmark as each additional dollar increase in spending reduces the shared savings payment by $0.75q$ dollars under the two-sided contract, and by $0.5q$ dollars under the one-sided contract. Additionally, if $y > b(1 + \bar{s})$, the penalty loss for each additional dollar increase in spending is $0.25q$ dollars under the two-sided contract, and zero under the one-sided contract.

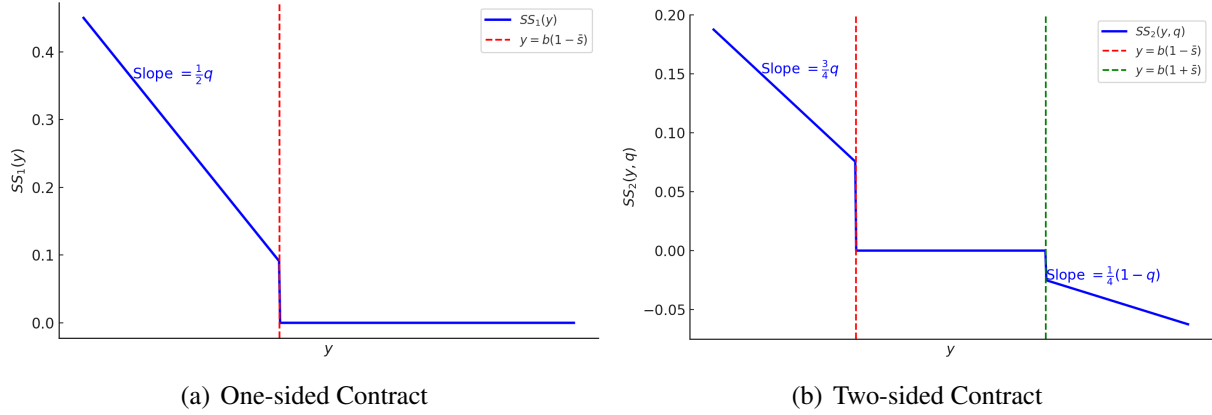


Figure 1: Shared Savings Payment

Notes: b denotes the benchmark, y is the ACO per-capita spending, q is the quality score, and $\underline{s}$ and $\bar{s}$ are the minimum savings rate and minimum loss rate respectively. Panel (a) shows the shared savings payments under the one-sided contract (with shared savings rate equal to $\frac{1}{2}$) as given by equation Equation 1. Panel (b) shows the shared savings payments under the two-sided contract (with shared savings rate equal to $\frac{3}{4}$ and shared loss rate equal to $\frac{1}{4}$) as given by Equation 2.

2.2 Benchmark

The ACO's participation in the MSSP is divided into Agreement Periods of three years, and each year under the MSSP is called Performance Year (PY). When an ACO joins MSSP, it enters a first Agreement Period (AP1) of three years with Medicare. The benchmark for AP1 is called the Historical Benchmark and is calculated based on the expenditure in the three Benchmark Years prior to joining MSSP. Two adjustments are applied to the ACOs' historical benchmark starting from the second Agreement Period. First, the ACO benchmark is rebased every three years based on the average spending of the previous three years. Secondly, a regional adjustment factor is applied to the benchmark in proportion to the difference between ACO rebased benchmark and average regional Medicare spending.

The ACO benchmark level and the benchmark updating rule are key design elements of the MSSP since the benchmark ultimately affects the amount of shared savings and the ACOs' participation decision. In this section, I explain in more detail the benchmark rebasement and regionalization, as well as how the benchmark is updated to account for changes in the ACOs' beneficiaries' health risk over time and variations of costs in the local healthcare market.

Historical Rebased Benchmark The historical benchmark is rebased at the start of each Agreement Period and is called the Rebased Historical Benchmark. This rebased benchmark is calculated

as the average spending in the three Performance Years of the Previous Agreement Period. The Performance Years used for calculating the rebased benchmark are called Benchmark Years (BY). Hence, the Historical Rebased Benchmark for the AP n uses the Performance Years of the previous AP $n - 1$ as Benchmark Years.

For instance, consider an ACO that joins the MSSP in 2015. The timeline of Agreement Periods, Performance Years and Benchmark years are illustrated in Figure 2. Let $b_{AP_n}^h$ denote the historical benchmark in Agreement Period n , and y_t the ACO per-capita spending in year t . In this case, years 2012 to 2014 are the Benchmark Years for the first AP, and the historical benchmark for the first AP is given by

$$b_{AP_1}^h = 0.6y_{2014} + 0.3y_{2013} + 0.1y_{2012}$$

and years 2015 to 2017 are the performance years for the first Agreement Period (PY AP1) and also Benchmark Years for second AP. The, the historical rebased benchmark for the second AP is given by

$$b_{AP_2}^h = \frac{1}{3}y_{2015} + \frac{1}{3}y_{2016} + \frac{1}{3}y_{2017}$$

and years 2018 to 2020 are the Performance Years of second AP.

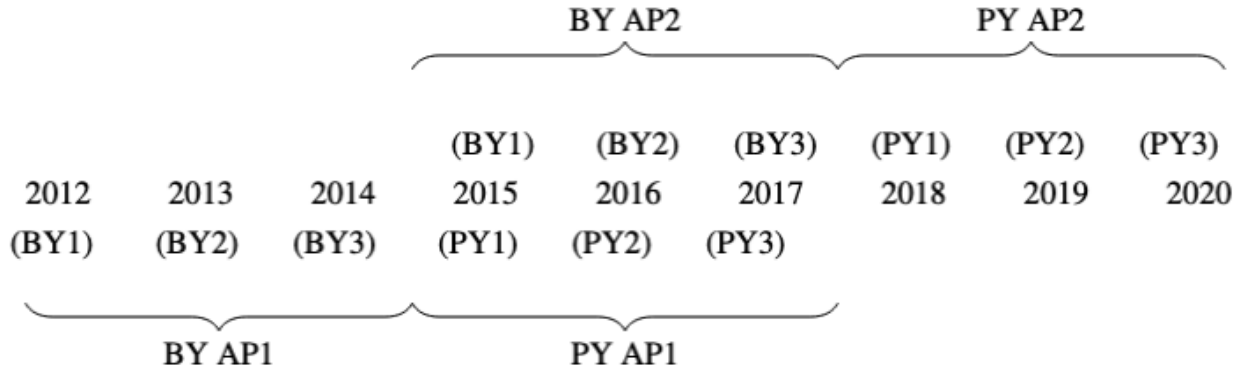


Figure 2: Timeline of MSSP Benchmark Rebasement

Notes: This Figure shows the timeline for an ACO that joins MSSP in 2015. AP denotes Agreement Period, PY denotes Performance Years and BY denotes Benchmark Years. Years from 2012 to 2014 are the BY for the first AP (AP1). Years from 2015 to 2017 are the PY of the first AP and also the BY of the second AP (AP2). Years from 2018 to 2020 are the PY of the second AP.

Updated Benchmark At the end of each performance year, the historical benchmark is adjusted to account for variations in patient health risk or prices in the local medical market between the

Benchmark Years and the current Performance Year. The risk and FFS price adjustments aim to recalculate the per-capita spending of the three Benchmark Years under the health risk and price conditions of the current Performance Year. The changes in the health status of the patient population are measured through the ratio of the average risk score in the current Performance Year and that of the most recent Benchmark Year. Similarly, variation in local medical market prices is measured using the trend factors representing the inflation in the national and regional FFS rates between the current Performance Year and the most recent Benchmark Year.

At the end of each performance year the historical (rebased) benchmark $b_{AP_n}^h$ is updated by the updated factor

$$\lambda_t = rr_t \cdot tf_t \quad (3)$$

where the rr_t denotes risk score ratio and tf_t is the FFS trend factor.¹ Hence, the updated benchmark that is relevant for the calculation of the shared savings payments is given by

$$b_t^u = b_{AP_n}^h \cdot \lambda_t \quad (4)$$

Going back to the example of the ACO joining the MSSP in 2015, the updated benchmark for performance years in the first AP is obtained by applying the update factor to the most recent Benchmark Year, which in this case is 2014. Similarly, the updated benchmark for performance years in the second AP is obtained by applying the update factor to the most recent Benchmark Year, which in this case is 2017.

Benchmark Regionalization Starting from the second Agreement Period, a regional adjustment factor is applied to the Rebased Historical Benchmark. Let y_t^R be Medicare's average per-capita risk-adjusted expenditure in the ACO's regional service area. The risk adjustment is used to account for the difference in the average health risk between the ACO's beneficiaries and the overall Medicare population of the region where the ACO's providers operate. The regional adjustment factor is calculated as a fraction w of the difference y_t^R and the ACO's expenditure y . The Regionalized Rebased Historical Benchmark hb_t^R is given by the sum of the Updated Rebased Historical Benchmark hb_t and the regional adjustment factor

$$b_t^{ur} = b_t^u + \underbrace{w(y_t^R - b_t^u)}_{\text{regional adjustment factor}} = (1 - w)b_t^u + wy_t^R \quad (5)$$

¹ $rr_t = \frac{r_t}{r_{\tau(t)}}$ where rr_t and $r_{\tau(t)}$ are the risk score in the current Performance Year and and most recent Benchmark Year.

Therefore, the Regionalized Rebased Historical Benchmark is a weighted average of the ACO's Rebased Historical Benchmark and the Medicare's average regional expenditure.

In the second Agreement period the weight assigned to the Medicare's average regional expenditure is

$$w = \begin{cases} 0.25 & \text{if } b_t^u > y_t^R \\ 0.35 & \text{if } b_t^u < y_t^R \end{cases}$$

Thus, if $b_t^u < y_t^R$ the adjustment is positive and the regionalized benchmark is larger than the rebased benchmark. Instead, if $b_t^u > y_t^R$ the adjustment is negative and the regionalized benchmark is lower than the rebased benchmark.²

2.3 Quality Score

An ACO's overall quality score is determined by the combination of 30-40 sub-measures of care quality. These sub-measures fall into the domains of "Patient/Caregiver Experience," "Care Coordination/Patient Safety," "Preventative Health," and "At-Risk Population." Patient/Caregiver Experience-related sub-measures are derived from survey responses and include the patient's doctor rating, access to specialists, and shared decision-making. "Care Coordination/Patient Safety" is a set of measures to evaluate (i) the ACO's effort at avoiding high-cost services like hospital unplanned admissions and readmissions, (ii) the ACO's utilization of tools to improve care coordination like Electronic Health Records (EHR). The "Preventative Health" domain measures the use of immunization, vaccination, and the use of screening to assess health conditions. The "At-Risk Population" is a set of sub-measures to evaluate the ACOs' effort to monitor and keep track of the health status of patients with chronic conditions.

For each sub-measure, quality points are assigned based on the level of performance relative to a benchmark measured using FFS data. The total points earned for measures in each domain will be summed and divided by the total points available for that domain to produce the overall quality score that will be used to determine the amount of shared savings (or shared losses).³

²The weight w for subsequent Agreement Periods is larger. See Appendix A for further details.

³See [Figure 10](#) in Appendix A

3 Data and Descriptive Evidence

The data for the analysis of this section and the estimation of the structural model are from MSSP ACO Public Use Files (2013-2022), MSSP Participant Lists, MSSP ACO Performance Year Results, and Number of ACO. Assigned Beneficiaries by County Public Use Files. In short, the data consists of ACO expenditures, benchmark expenditures, quality scores, contract choice, assigned beneficiary demographics, and various participant and provider statistics.

Little public information is available on the characteristics of specific ACO participants. I complement this data with other data sources:

- (i) the Physician Compare Database, which contains detailed information on the providers' characteristics (specialty, experience, location, beneficiaries risk score, hospital affiliations, EHR usage).
- (ii) the American Hospital Association (AHA) Survey Database which allows to have information on the degree of integration between hospitals and physicians' organizations.

3.1 Descriptive Evidence

Table 2 in Appendix B shows some summary statistics on ACOs' characteristics and financial performance. The number of ACOs has increased significantly over time, and currently, there are more than 500 active ACOs in the MSSP. Moreover, around a hundred new ACOs enter the MSSP every year, and fifty active ACOs exit the MSSP every year ⁴. The vast majority of the ACOs opted for the one-sided contract, but the share of ACOs in the two-sided contract has increased significantly over the last few years.

There is a large variation across ACOs in the number of providers and the number of assigned beneficiaries, and the mean of these variables appear to be increasing over time. The average risk score of the ACO's assigned beneficiaries is above one until 2016, and around one from 2017 onward due to the exit of some ACOs with high risk score.⁵ The benchmark per-capita is, on average, between 10 to 11 thousand dollars, which is close to the mean per-capita expenditure of the overall Medicare population.

On average, only a third of the ACOs have been eligible for shared savings, but the trend has been consistently increasing over time. However, the spending reduction generated by the ACOs has been very limited. For example, in 2015, the total savings across all ACOs was approximately

⁴See Figure 11 in Appendix A.

⁵The average risk score of the Medicare population is 1 by construction.

\$430 million, which represents a 0.6% decrease in Medicare spending. This can be seen from the average savings rate, which is around 1%, but it has increased steadily over time.

The ACO does not have perfect control over the average per-capita spending since the latter depends on variations in patients' demand and health status and also because each ACO provider remains independent and free to choose the level of spending that maximizes its own profits. However, the ACO can affect per-capita spending through its efforts to generate shared savings. In the structural model, I will use the ACO level quality score $q \in [0, 1]$ as a measure of the ACO effort at reducing expenditure through improved care coordination, preventative care, and care monitoring.

Table 3 in Appendix B shows the impact of quality on the level of utilization and spending of several high-cost services after controlling for patient's risk scores and other ACO characteristics. A one standard deviation increase in the quality score leads to (i) a 1.8% reduction in the Hospital Readmissions rate, (ii) a 2.5% reduction in Emergency Department Visits that Lead to Hospitalization, (iii) a 1.9% reduction in per-capita on inpatient (IP) spending and 3.5% reduction in outpatient (OP) spending, and (iv) a 4.1% reduction in the per-capita spending on Ambulance Services.

3.2 Selection on Level

ACOs have incentives to select into MSSP on both levels. Following Einav et al., 2022, selection on levels means that ACOs have a larger incentive to select in if their average spending, holding behavior fixed at what it would be under FFS, would be below their benchmark.

The benchmark used in the first Agreement Period is the historical benchmark, which is calculated as the average of the ACOs' spending in the three years before joining the MSSP. Therefore, as long as the historical benchmark is not too far from FFS spending, ACOs do not select on level. Indeed, if the ACO keeps spending at the FFS level in the first Agreement Period, the amount of shared savings is approximately zero.

However, selection on level can occur when the regional adjustment factor is applied to the ACO benchmark starting from the second Agreement Period. As described in Section 1, when the ACO's spending is lower than the Medicare average regional spending, a positive regional adjustment factor is applied to the ACO's historical rebased benchmark. Hence, starting from the second agreement period, ACOs with a positive regional adjustment factor might be able to obtain shared savings without exerting effort to reduce spending.

I test for the presence of selection on level by estimating the impact of the regional adjustment factor on the probability of leaving the MSSP. Table 4 in Appendix B shows the results of a logistic

regression of the exit decision on a regional adjustment factor dummy variable and a set of control variables. The odds ratio of 0.33 indicates that for those ACOs with a positive regional adjustment factor, the odds of leaving the MSSP are about one third compared to that of ACOs with negative regional adjustment factor. Thus, ACOs with a positive regional adjustment factor are more likely to join and stay in the MSSP, which indicates the presence of selection on levels.

3.3 Potential ACOs

The main objective of this paper is to study the impact of the MSSP benchmarking rules on the MSSP outcomes. Hence, I need to estimate how ACOs' participation changes under alternative MSSP benchmarking rules. Under different benchmarking rules, some ACOs may choose not to participate or exit the MSSP, while new ACOs may enter the program. However, the new potential ACOs that could form under alternative rules are not observed in the data, as we can only see the ACOs that actually participated in the MSSP. Therefore, it is necessary to construct these potential ACOs using alternative data sources.

Since ACOs are groups of independent providers that form a joint venture to coordinate the care of a patient population, I use the Physician Shared Patients Patterns Data to identify networks of providers that resemble actual ACOs. This database records the number of unique patients shared between each pair of providers, providing a measure of how closely providers are connected through their patient population. We construct providers' networks, where nodes represent providers, and edges represent the number of patients they share. The strength of each edge is determined by the number of unique shared patients, which serves as a measure of the degree of collaboration between providers.

To identify groups of providers that resembles the actual ACOs, we apply a community detection algorithm to this network. Specifically, we use the Louvain clustering method, which is a widely used algorithm for detecting strongly interconnected groups in large networks. This approach allows us to detect natural provider groupings that resemble actual ACOs.

To validate this approach, I compare the identified providers' networks with actual ACOs as reported in the MSSP Participant Lists database. If the inferred networks closely match the actual ACOs, we can conclude that the algorithm is effective at capturing ACO-like structures. This gives us confidence that it can also identify the potential ACOs that would have formed under alternative MSSP rules. These potential ACOs are then used in counterfactual simulations to analyze how changes in benchmarking rules impact ACOs' participation decisions and the overall savings generated by the MSSP.

4 Theoretical Model

In this section, I first discuss the challenges of modeling the utility of an ACO and present the modeling assumptions. Secondly, I derive the optimality conditions under FFS and MSSP in a static setting. Finally, I present a simplified version of the dynamic model and show the effect of the benchmark rebasement on ACOs' behavior.

4.1 The utility of an ACO

A common assumption in the health economics literature is to model providers as deriving utility from patients' health benefits and financial profits. Since ACOs are joint ventures created by independent healthcare providers, I also model the utility of an ACO as depending on patients' health benefits and profits.⁶ Patients' health benefits enter the ACO's utility because norms of professional conduct require providers to advocate for the patients' interest and because ACOs are monitored on the health outcomes and the quality of Patient-Caregiver experience. Following the paper by [Gaynor et al. \(2004\)](#) on providers' behavior within Health Maintenance Organizations (HMO), I assume that patients' health benefits depend on the extent to which the ACO per-capita spending y deviates from some reference level healthcare expenditure y^r . I conceive of y^r as representing the level of spending that maximizes patients' health. The level of y^r varies among ACOs and over time for the same ACO as it depends on the utilization and health status of the patient population.

The financial profits of the ACO depend on both the shared savings payments and the Medicare FFS reimbursement since each ACO provider is still paid according to the standards FFS system. Thus, the main challenge is to model how the FFS profit functions of several providers enter the ACO's utility. Secondly, the average ACO per-capita spending is the result of the claims submitted by multiple independent providers. Therefore, the ACO does not have direct control over the average ACO per-capita spending. The ACOs' activity revolves around enhancing the efficiency of healthcare delivery through improved care coordination, preventative care, and care monitoring, which in turn impacts the average ACO per-capita spending and the shared savings obtained by the ACO. Hence, another challenge is to model how the ACO's activities affect the average ACO per-capita spending and the cost associated with those activities.

Since this paper aims to study how the benchmarking rules affect the ACOs' participation in the MSSP and their effort to reduce spending, I abstract from the complex interaction between the ACO providers and model each ACO as a single agent. I assume that ACOs exert effort

⁶See McGuire (2000) for a review of the general literature on models of physician behavior.

in care coordination, preventative care and care monitoring as measured by the quality score q . Furthermore, to capture the fact that the ACO does not have direct control over the average ACO per-capita spending, I model the latter as a function of the reference spending y^r and the quality score $q \in (0, 1]$

$$y(q) = y^r q^{-\alpha} \quad (6)$$

where $-\alpha$ is the elasticity of spending with respect to quality. I also assume the cost of the quality score is heterogeneous across ACOs and varies based on ACOs' observable characteristics including the risk score of the ACOs beneficiaries.

Finally, to capture the role played by the FFS incentives, I follow [Aswani et al. \(2019\)](#) assumption that the FFS per-capita spending enters positively in the ACO's utility as $m \cdot y(q)$ where $m \in [0, 1]$ can be interpreted as the profit margin on the FFS rates. I also assume that the ACO existed before it participated in the MSSP. Since the historical benchmark of the first AP is computed using the spending of the ACO providers in the three years before the first performance year, it is reasonable to assume that some degree of care coordination among healthcare providers also occurs without the MSSP incentives.

Let d_j denote the ACO j discrete choice among staying under the FFS system ($d_j = 0$) or choosing to join the MSSP under the one-sided ($d_j = 1$) or two-sided contract ($d_j = 2$). The shared savings payments $SS(d_j, y(q_j), q_j)$ are given by

$$SS(d_j, y(q_j), q_j) = \begin{cases} 0 & \text{if } d_j = 0 \\ SS_1(y(q_j), q_j) & \text{if } d_j = 1 \\ SS_2(y(q_j), q_j) & \text{if } d_j = 2 \end{cases} \quad (7)$$

where $SS_1(y(q_j), q_j)$ and $SS_2(y(q_j), q_j)$ are given by [Equation 1](#) and [Equation 2](#) respectively.

As described in the Section 2, at the end of each performance year the historical benchmark is updated by the update factor λ_t ([Equation 3](#)) to reflect changes in risk score and FFS rates. However, the update factor is not revealed by CMS until the end of the performance year. Therefore, ACOs make their decisions under uncertainty about the update factor used to compute the updated benchmark. I assume that the updated factor is normally distributed $\lambda \sim N(\mu_\lambda, \sigma_\lambda)$. The expected value of the shared savings payments under the one-sided contract ($d_j = 1$) is therefore given by

$$\mathbb{E}_\lambda [SS(1, y(q_j), q_j)] = \frac{1}{2} q_j \left[\left(b^h \cdot \mu_{\lambda_1} - y(q_j) \right) \left(1 - \Phi(\lambda_1(q_j)) \right) + \sigma_\lambda \cdot b^h \cdot \phi(\lambda_1(q_j)) \right]$$

and under the two-sided contract ($d_j = 2$)

$$\mathbb{E}_\lambda \left[SS(2, y(q_j), q_j) \right] = \frac{3}{4} q_j \left[\left(b^h \cdot \mu_{\lambda_1} - y(q_j) \right) \left(1 - \Phi(\lambda_1(q_j)) \right) + \sigma_\lambda \cdot b^h \cdot \phi(\lambda_1(q_j)) \right] + \frac{1}{4} (1 - q_j) \left[\left(b^h \cdot \mu_\lambda - y(q_j) \right) \Phi(\lambda_2(q_j)) - \sigma_\lambda \cdot b^h \cdot \phi(\lambda_2(q_j)) \right]$$

where b^h is the historical benchmark, $\lambda_1(q_j) = \sigma_\lambda^{-1} \left(\frac{y}{b^h(1-s)} - \mu_\lambda \right)$, $\lambda_2(q_j) = \sigma_\lambda^{-1} \left(\frac{y}{b^h(1+s)} - \mu_\lambda \right)$ and Φ and ϕ are the standard normal cdf and pdf respectively. The shared savings contract induces spending reductions by making higher-quality choices more profitable through bonus payments that depend on quality. Notice that higher quality increases the expected shared savings payments both directly since the shared savings rate is multiplied by the quality score q and indirectly through lower spending relative to the benchmark.

A major barrier that ACOs face with respect to participating in the MSSP is the significant investment that must be made to improve healthcare delivery efficiency (Haywood and Kosel, 2011). For example, an ACO may need to invest in new information technology to better coordinate care for its patients (Moore and Coddington, 2011) or invest in increasing the quality of care delivered so that unnecessary and costly services are minimized (Anderson and Steinberg 1984). In particular, the MSSP requires ACO providers to have a functioning Electronic Health Record system (EHR) that facilitates the exchange of information and coordination among providers. Therefore, ACOs pay an entry cost when they join the MSSP, and I assume the value of the entry cost varies across ACOs.

Thus, ACO j financial profits are given by the sum of the FFS incentives and the MSSP incentives minus the cost of the quality score and the entry cost. The financial profits $\pi_j(d_j, q_j)$ of choosing option d_j and quality score q_j are given by

$$\pi_j(d_j, q_j) = m \cdot y(q_j) + \mathbb{E}_\lambda \left[SS(d_j, y(q_j), q_j) \right] - C_j(q_j) - K_j \cdot 1(d_j > 0) \quad (8)$$

The expected utility of an ACO with a patient population of size N_j is given by

$$\mathbb{E}_\lambda [u_j(d_j, q_j)] = N_j \left[H(y(q_j), y_j^r) + \pi_j(d_j, q_j) \right] + \varepsilon_j^{d_j} \quad (9)$$

where $H(y(q_j), y_j^r)$ are the patients health benefits, $\pi_j(d_j, q_j)$ are the financial incentives given by Equation 8 and $\varepsilon_j^{d_j}$ is a contract specific utility shock. Notice that $H(\cdot)$, $\pi_j(\cdot)$ and K_j are expressed in per-capita terms.

In the empirical model I will assume that the cost is quadratic in quality score

$$C_j(q_j) = c_j q_j^2$$

where c_j is an ACO-specific cost parameter. As described above, I model patients health benefits as the deviation between the per-capita spending $y(q_j)$ and the reference spending y_j^r

$$H(y(q_j), y_j^r) = -\eta \left(y(q_j) - y_j^r \right)^2$$

where η is a positive parameter common across ACOs. Therefore, the utility specification that I will use for the estimation is

$$\begin{aligned} \mathbb{E}_\lambda[u_j(d_j, q_j)] = & N_j \left[-\eta \left(y(q_j) - y_j^r \right)^2 + m \cdot y(q_j) + \right. \\ & \left. \mathbb{E}_\lambda[SS(d_j, y(q_j), q_j)] - C_j(q_j) - K_j \cdot 1(d_j > 0) \right] + \varepsilon^{d_j} \end{aligned} \quad (10)$$

The quality score affects the ACOs' utility directly through the cost function and also indirectly through spending. Higher quality score decreases spending, which in turn impacts (i) negatively FFS profits and (ii) positively MSSP expected shared savings. Patients' health benefits are also affected by the quality score, but the sign of the impact depends on whether $y(q_j)$ is above or below y_j^r .

Quality score under FFS and MSSP Suppose that the ACOs do not take into account the benchmark rebasement and regionalization, and assume that they simply maximize the current year utility as given by (Equation 10). Conditional on choosing to stay under FFS ($d_j = 0$), the optimality condition for the quality score is given by

$$m \frac{\partial y}{\partial q_j} - 2c_j q_j - 2\eta \left(y(q_j) - y_j^r \right) \frac{\partial y}{\partial q_j} = 0$$

The optimal quality score is set to balance the loss due to decrease in FFS revenue and higher quality costs with the positive change in patients benefits.

In Figure 3 the ACO's utility under FFS as defined by Equation 10 under $d_j = 1$ is represented by the blue line. The red line shows the ACO spending as function of the quality score as defined by Equation 6. At the optimal quality score q^F (vertical dashed line), the difference between the FFS spending y^F (upper horizontal dotted line) and the reference spending y^r (lower horizontal dotted line) represents the amount of overspending under FFS. It can be showed that the overspending

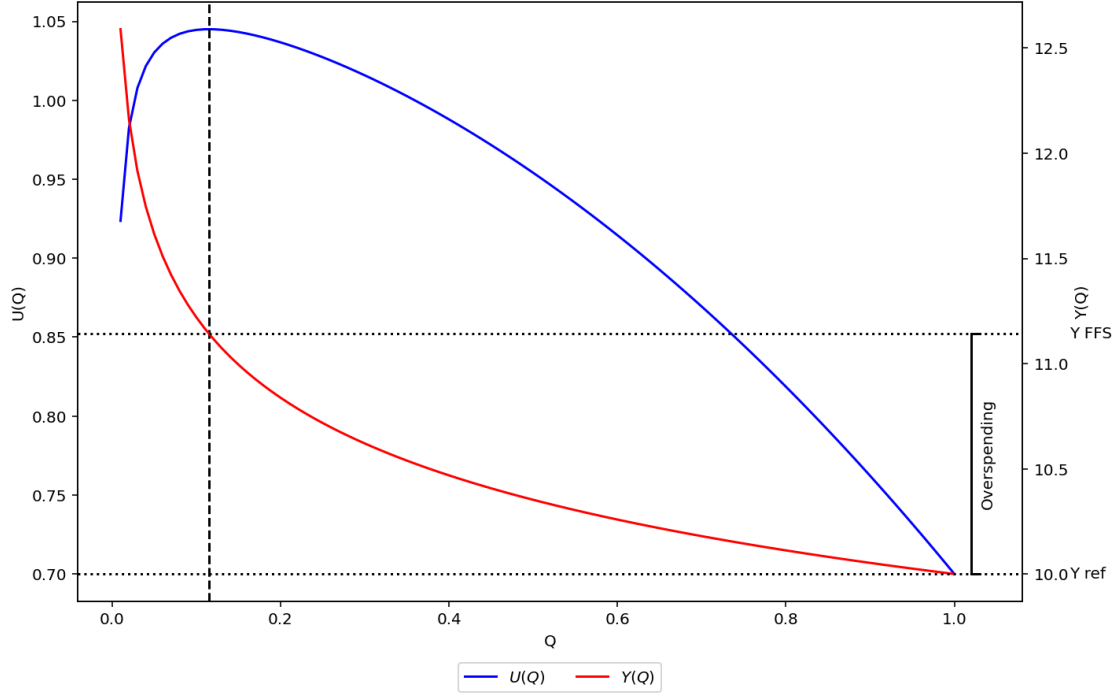


Figure 3: Utility and spending under FFS

Notes: The x-axis shows the quality score $q \in (0, 1]$. The lower horizontal dotted line indicates the reference spending value $y^r = 10$. The blue line plotted on the left hand side shows the ACO's utility under FFS as defined by Equation 10 under $d_j = 0$ and $y^r = 10$. The red line plotted on the right hand side shows the ACO spending as function of the quality score as defined by Equation 6. The vertical dashed line indicates the optimal quality score under FFS q^F , and the upper horizontal dotted line indicates the FFS spending value that corresponds to $y^F = y^r (q^F)^{-\alpha}$. The vertical difference between y^F and y^r represents the amount of overspending under FFS.

depends positively on the FFS profit margin m and the cost parameter c , and negatively on the elasticity of spending with respect to quality $-\alpha$ and the weight parameter on the patients health benefits η .

Under FFS, the only incentive to choose a positive quality score is represented by the patients' health benefits term. In other words, the quality score is used to control the level of overspending relative to y_r . We can think of the MSSP objective as setting incentives to induce the ACOs to reduce spending to a level close to y^r through better care coordination and preventative care (i.e., higher quality score). In particular, the shared savings contract incentivizes ACOs to increase quality score and reduce spending.

Thus, under the MSSP ($d_j > 0$), the optimal q_j satisfies the first order condition (FOC)

$$m \frac{\partial y}{\partial q_j} - 2c_j q_j - 2\eta(y(q_j) - y_j^r) \frac{\partial y}{\partial q_j} + \frac{\partial \mathbb{E}(SS(d_j, y(q_j), q_j))}{\partial q_j} = 0$$

Notice that the FOC under MSSP is identical to the FOC under FFS except for the derivative of the expected shared savings with respect to the quality score. Since the sign of this derivative is positive, the optimal quality under MSSP q^M is larger than q^F .⁷ Therefore, the MSSP contract induces the ACOs to reduce the overspending relative to the level that would be observed under FFS.

Figure 4 shows the ACO's utility under MSSP (blue line) and the ACO spending (red line) as function of the quality score. Suppose the benchmark is set to be equal to the counterfactual FFS spending y^F . At the optimal quality score q^M (vertical dashed line), the difference between the MSSP spending y^M (lower horizontal dotted line) and the FFS spending y^F (upper horizontal dotted line) represents the spending reduction (savings) relative to the benchmark. It can be showed that the savings depends positively on the negatively on the FFS profit margin m and the cost parameter c , and positively on the elasticity of spending with respect to quality $-\alpha$.

4.2 Dynamic Model: Ratchet Effect

In this section, I investigate the effect of benchmark rebasement on the ACO's participation choice and quality score choices. Since participation in the MSSP is voluntary, each year, ACOs choose whether to stay under the FFS system or participate in the MSSP. However, I assume that ACOs exit the MSSP cannot re-enter and stay under FFS indefinitely. This assumption is made for analytical reasons detailed in Section 5 and is justified by the fact that I do not observe any ACO re-entering the MSSP after exit.

Conditional on the participation choice, ACO j optimally chooses the quality score, considering how spending in the current performance year affects the rebased benchmark in the next Agreement Period. In this setting, both the participation and quality score are dynamic choices. Therefore, each year, the ACO chooses d_j and q_j to maximize the current utility and the expected discounted value of future utility.

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$$\frac{\partial \mathbb{E}(SS(d_j, y(q_j), q_j))}{\partial q_j} = \underbrace{\frac{1}{q_j} \mathbb{E}(SS(d_j, y(q_j), q_j))}_{>0} + \underbrace{\frac{\partial \mathbb{E}(SS(d_j, y(q_j), q_j))}{\partial y}}_{<0} \underbrace{\frac{\partial y(q_j)}{\partial q_j}}_{<0} > 0$$

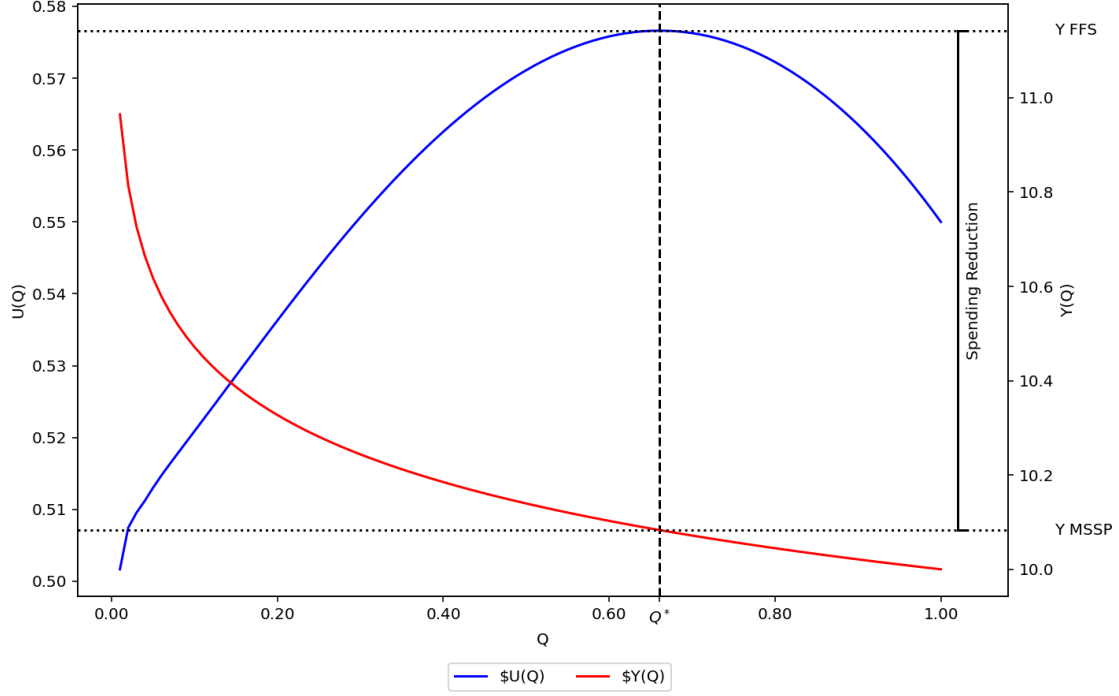


Figure 4: Utility and spending under MSSP

Notes: The x-axis shows the quality score $q \in (0, 1]$. The upper horizontal dotted line indicates the spending value under FFS. The blue line plotted on the left hand side shows the ACO's utility under FFS as defined by Equation 10 under $d_j = 1$ and $y^r = 10$. The red line plotted on the right hand side shows the ACO spending as function of the quality score as defined by Equation 6. The vertical dashed line indicates the optimal quality score under MSSP q^M , and the lower horizontal dotted line indicates the MSSP spending value that corresponds to $y^M = y^r(q^M)^{-\alpha}$. The vertical difference between y^F and y^r represents the amount of overspending under FFS.

Since this section aims to illustrate the effect of the benchmark rebasement, I will consider a simplified version of the dynamic model. I make four assumptions for ease of analytical tractability. I assume that (i) the initial benchmark is given by the FFS spending in the year before entry, (ii) the benchmark is rebased every year, (iii) the regional adjustment is applied every year, (iv) ACOs can only choose between FFS and one-sided MSSP contract and (v) there are only two state variables, the benchmark b_t and the risk score r_t . These simplifying assumptions will be removed in the empirical model of Section 5.

For ease of notation, I will omit the subscript j . Let y_t^0 and y_t^1 denote the ACOs per-capita spending in year t under FFS and MSSP respectively, and y_t^R the Medicare regional average per-capita spending in year t . Under assumptions (i)-(iii), the historical benchmark b^h evolves over

time according to the following rule

$$b_{t+1}^h = \begin{cases} y_t^0 & \text{if } d_t = 0 \\ (1-w)y_t^1 + wy_t^R & \text{if } d_t = 1 \end{cases} \quad (11)$$

where w is the weight on the regional adjustment factor.

Let $v_t^0(b_t, r_t)$ and $v_t^1(b_t, r_t)$ denote the choice-specific value function of FFS and MSSP respectively, given the current period state variables (b_t, r_t) . An ACO participates in the MSSP ($d_t = 1$) if the value of participation is larger than that of staying under FFS ($d_t = 0$)

$$v_t^1(b_t^h, r_t) + \epsilon_t^1 > v_t^0(b_t^h, r_t) + \epsilon_t^0$$

where ϵ_t^0 and ϵ_t^1 are the FFS and MSSP specific extreme value shocks.

Since ACOs are forward-looking agents, they take into account the fact that higher quality score in the current year leads to lower spending and, in turn, lower benchmark in the next year. Hence, the choice-specific value function of participating in the MSSP satisfies the Bellman equation

$$v_t^1(b_t, r_t) = \mathbb{E}_\lambda \left[u(1, q_t^1 | b_t^h, r_t) \right] + \beta \mathbb{E}_r \left[V_{t+1}(b_{t+1}^h, r_{t+1}) \right] \quad \text{s.t.} \quad b_{t+1}^h = (1-w)y(q_t^1) + wy_t^R$$

where q_t^1 denotes the optimal quality score under MSSP, and V_{t+1} is the continuation value function. Notice the expected value of the utility is given by [Equation 10](#), and the expectation of V_{t+1} is on the distribution of next year risk score.

The first order condition of this problem is given by

$$\frac{\partial \mathbb{E}_\lambda u(1, q_t | b_t^h, r_t)}{\partial q_t} + \beta \mathbb{E}_r \left[P_{1t+1}(b_{t+1}^h, r_t) \frac{\partial v_{1t+1}(b_{t+1}^h, r_{t+1})}{\partial b_{t+1}^h} \right] \frac{\partial b_{t+1}^h}{\partial y_t} \frac{\partial y_t}{\partial q_t} = 0 \quad (12)$$

where $P_{1t+1}(b_{t+1}^h, r_t)$ is the probability of participating in the MSSP in the next period ⁸. This expression contains the derivative of the unknown value function v_{t+1}^1 . Taking the derivative of

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$$P_{1t+1}(b_{t+1}^h, r_t) = \frac{\exp \left(v_{t+1}^1(b_{t+1}^h(y_t^M), r_{t+1}) \right)}{\exp \left(v_{t+1}^1(b_{t+1}^h(y_t^M), r_{t+1}) \right) + \exp \left(v_{t+1}^0(b_{t+1}^h(y_t^F), r_{t+1}) \right)}$$

$v_t^1(b_t^h, r_t)$ with respect to b_t^h and applying the Envelope Theorem, it can be shown that

$$\begin{aligned} \frac{\partial v_t^1}{\partial b_t^h}(b_t, r_t) &= \underbrace{\frac{\partial \mathbb{E}_\lambda u_t}{\partial q_t}(1, q_t^1 | b_t^h, r_t) + \beta \mathbb{E}_r \left[P_{1t+1}(b_{t+1}^h, r_t) \frac{\partial v_{1t+1}(b_{t+1}^h, r_{t+1})}{\partial b_{t+1}^h} \right] \frac{\partial b_{t+1}^h}{\partial y_t} \frac{\partial y_t}{\partial q_t}}_{= 0 \text{ from FOC}} + \frac{\partial \mathbb{E}_\lambda u_t}{\partial b_t^h}(1, q_t^1 | b_t^h, r_t) \\ &= \frac{\partial \mathbb{E}_\lambda u_t}{\partial b_t^h}(1, q_t^1 | b_t^h, r_t) \end{aligned} \quad (13)$$

Substituting this equation for $t + 1$ in Equation 12, we obtain the Euler Equation for quality:

$$\frac{\partial \mathbb{E}_\lambda u_t}{\partial q_t}(1, q_t^1, b_t^h, r_t) + \beta \mathbb{E}_r \left[P_{1t+1}(b_{t+1}^h, r_t) \frac{\partial \mathbb{E}_\lambda u_{t+1}}{\partial b_{t+1}^h}(1, q_{t+1}^1, b_{t+1}^h, r_{t+1}) \right] \frac{\partial b_{t+1}^h}{\partial y_t} \frac{\partial y_t}{\partial q_t} = 0 \quad (14)$$

Since higher quality score in year t leads to lower spending and, in turn, a lower benchmark and lower utility in the year $t + 1$, the second term of the Euler Equation is negative

$$\underbrace{\frac{\partial \mathbb{E}_\lambda u_{t+1}}{\partial b_{t+1}^h}}_{>0} \underbrace{\frac{\partial b_{t+1}^h}{\partial y_t}}_{>0} \underbrace{\frac{\partial y_t}{\partial q_t}}_{<0} \leq 0$$

This implies that the first term of the Euler equation is positive

$$\frac{\partial \mathbb{E}_\lambda u_t(1, q_t^1, b_t^h, r_t)}{\partial q_t} \geq 0$$

Therefore, at optimum, the marginal gain in tomorrow's utility from having a higher benchmark has to be equal to the marginal loss in today's utility from keeping the quality score below the level that would maximize a static utility. In other words, the Euler Equation implies that the optimal quality score is lower than the quality score that would be optimal without benchmark rebasement. From Medicare's perspective, it may be ideal for ACOs to act myopically, that is, to achieve the largest possible spending reduction each year. However, due to the MSSP rebasement mechanism, ACOs may find it strategically advantageous to reduce their efforts (and generate smaller savings) in earlier years to secure more favorable benchmarks in subsequent periods. This behavior is called the ratchet effect. The extent of the ratchet effect is controlled by the weight of the regional adjustment factor. If w is set to one, the level of spending in the current year has no impact on the next year's benchmark the benchmark is fully regionalized.

$$\frac{\partial b_{t+1}^h}{\partial y_t} = 1 - w = 0 \quad \text{if} \quad w = 1$$

Therefore, if $w = 1$, the optimal level of quality score is set to maximize the static utility

$$\frac{\partial \mathbb{E}_\lambda u_t(1, q_t^1, b_t^h, r_t)}{\partial q_t} = 0$$

and there is no ratchet effect. Indeed, the regional adjustment factor ([Equation 5](#)) was introduced starting from 2017 as a mechanism to incentivize ACOs to sustain spending reductions over time and mitigate the ratchet effect.

In a seminal paper, [Shleifer \(1985\)](#) highlights how regulated local monopolies often lack incentives to reduce costs in environments where regulators set prices based on the firms' own costs. To address this issue, the author proposes yardstick competition, where regulators set a firm's prices based on the costs of comparable firms. Benchmark regionalization can be seen as an attempt to incorporate elements of yardstick competition into MSSP benchmarking. However, while fully regionalized benchmarks can help prevent the ratchet effect, they may also discourage ACO participation, especially if some ACOs find such benchmarks unattainable. To balance these concerns, the MSSP blends the rebased benchmark with the spending patterns of Medicare providers in the same service area, adjusting for differences in patient population health risks. This approach aims to regulate the strategic delay in spending reduction of forward-looking ACOs, while maintaining participation incentives.

As discussed in Section 3, benchmark regionalization is not an ideal solution because it can result in adverse selection. This suggests the need for an alternative benchmarking rule that better balances the trade-offs between rebasement and regionalization. Ideally, such a rule would use benchmark regionalization to mitigate the ratchet effect while also addressing the issue of adverse selection. In Section 6, I will introduce and evaluate the impact of an alternative benchmarking rule called Conditional Regionalization. This approach modifies the current rule by making the application of the regional adjustment factor conditional on the ACO having generated savings in the previous AP. ACOs that would adversely select in under the current MSSP, have now little incentive to participate, as they cannot earn shared savings without reducing spending relative to the FFS level. At the same time, Conditional Regionalization still mitigates the ratchet effect by blending the rebased benchmark with the Medicare regional average spending, as in the current MSSP framework.

5 Empirical Model

This section presents the empirical model and shows the estimation results. First, I relax the simplifying assumptions (i)–(v) of Section 4.2 and extend the dynamic model to the institutional setting discussed in Section 2. I also revisit the benchmarking rules outlined in Section 2 to illustrate the evolution of the ACOs’ state variable over time. Secondly, I present the ACOs’ value function and discuss the identification of the ACOs primitives. Finally, I describe the estimation strategy and show the estimation results.

Each year t , ACO j makes two decisions: the participation decision $d_{jt} \in \{0, 1, 2\}$ and the quality score $q_{jt} \in (0, 1]$. The choice set for the participation decision consists of three options: (i) the standard Fee-for-Service regime $d_{jt} = 0$, (ii) the one-sided MSSP contract $d_{jt} = 1$ and (iii) the two-sided MSSP contract $d_{jt} = 2$. The ACOs’ participation in the MSSP is divided into Agreement Periods of three Performance Years each. Let AP_t and PY_t denote the agreement period and the performance year, respectively. In $t = 0$, all ACOs are under FFS. ACOs that are under FFS can choose to stay under FFS or participate in the MSSP. ACOs participating in the MSSP can choose to remain in the MSSP or leave the program. Moreover, the decision to leave the MSSP is permanent, implying that ACOs cannot re-enter the MSSP after exiting.

As described in Section 2, the historical benchmark at the start of the first AP is given the weighted average spending in the three years prior to joining the MSSP

$$\bar{y}_{jt} = 0.1y_{jt-1} + 0.3y_{jt-2} + 0.6y_{jt-3}$$

and at the start of each subsequent AP the historical benchmark is rebased using the average spending in the previous AP. Hence, ACOs need to keep track of the evolution of the average spending over time. Let $\tilde{y}_{jt}$ denote the rolling average spending of the current AP before the current PY spending is realized. After the current PY spending y_{jt} is realized, the next year rolling average spending is given by

$$\tilde{y}_{jt+1} = \begin{cases} y_{jt} & \text{if } PY_{jt} = 1 \\ \frac{1}{2}y_{jt} + \frac{1}{2}\tilde{y}_{jt} & \text{if } PY_{jt} = 2 \\ \frac{1}{3}y_{jt} + \frac{2}{3}\tilde{y}_{jt} & \text{if } PY_{jt} = 3 \end{cases}$$

Hence, the historical rebased benchmark at the start of each AP is given by

$$b_{jt}^h = \begin{cases} \bar{y}_{jt} & \text{if } PY_{jt} = 1 \text{ and } AP_{jt} = 1 \\ \tilde{y}_{jt} & \text{if } PY_{jt} = 1 \text{ and } AP_{jt} > 1 \end{cases}$$

Moreover, from Section 2 and 4 we know that the ACO also has to take into account the evolution of the risk score and FFS trend factor since at the end of each performance year PY the historical rebased benchmark is updated as follows

$$b_{jt}^u = b_{jt}^h \cdot \lambda_{jt}$$

where λ_{jt} is the update factor of ACO j as defined in Equation 3.

Since starting from the second AP the benchmark is regionalized, the ACO also keeps track of the average Medicare expenditure in its regional service area $\bar{y}_{jt}^R$. The regionalized historical (rebased) benchmark is a weighted average of b_{jt}^u and $\bar{y}_{jt}^R$

$$b_{jt+1}^{ru} = (1 - w)\bar{y}_{jt} + w\bar{y}_{jt}^R$$

where w is the weight on the regional expenditure.⁹

Therefore, the state variables at time t that are observable by ACO j and the researcher before ACO j makes the participation decision are given by

$$\Omega_{jt} = \{AP_{jt-1}, PY_{jt-1}, b_{jt-1}^h, \tilde{y}_{jt}, y_{jt}^R, \lambda_{jt-1}, N_{jt}, a_j\}$$

where N_{jt} is the number of assigned beneficiaries, a_j is the ACO type. Finally, the ACOs' program choice is also affected by unobservable idiosyncratic shocks ε_{it}^d for $d = \{0, 1, 2\}$ which I assume to be Type 1 Extreme Value distributed.

Timing of the model The timing of actions and information revelation within each period is as represented in the timeline in Figure 5:

- (i) At the beginning of the period, the ACO considers the historical rebased benchmark b_{jt}^h and update factor λ_{jt-1} , and the rolling average spending $\tilde{y}_{jt}$ and regional average spending y_{jt}^R , and observes the realization of the current period idiosyncratic shock ε_{jt}^d for $d \in \{0, 1, 2\}$.
- (ii) The ACO forms expectation on the evolution of the update factor and reference spending. Given these expectation and ε_{jt}^d , the ACO makes the program participation choice $d_{jt} \in \{0, 1, 2\}$.

⁹The value of w depends on the sign of the regional deviation and it varies across Agreement Periods. See Appendix for details.

(iii) AP_{jt-1} and PY_{jt-1} are updated with the following rules:

$$PY_{jt} = \begin{cases} PY_{jt-1} + 1 & \text{if } PY_{jt-1} < 3 \text{ and } d_{jt} = 1 \\ 1 & \text{if } PY_{jt-1} = 3 \text{ and } d_{jt} = 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$AP_{jt} = \begin{cases} AP_{jt} & \text{if } PY_{jt-1} < 3 \text{ and } d_{jt} = 1 \\ AP_{jt} + 1 & \text{if } PY_{jt-1} = 3 \text{ and } d_{jt} = 1 \\ 0 & \text{otherwise} \end{cases}$$

(iv) The year t value of y_{jt}^r .

(v) The ACO optimally chooses the quality score q_{jt}^d conditional on the program choice d , and the ACO spending level is realized according to [Equation 6](#).

(vi) The year t value of the updated factor λ_{jt} is realized, and the benchmark is updated b_{jt}^u . Finally, the benchmark is updated, shared savings payments are made and the current year utility is realized.

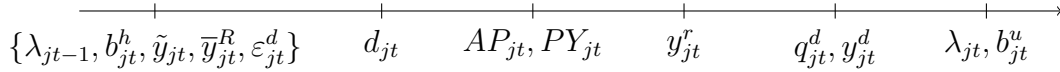


Figure 5: Timeline of the model

Value function In each period the ACOs choose the program $d_{jt} \in \{0, 1, 2\}$ and quality score $q_{jt} \in (0, 1]$ that gives the highest expected lifetime utility. For a given program d , the choice specific value function can be written as:

$$v_j^d(q_{jt}^d, \Omega_{jt}) + \varepsilon_{jt}^d = \mathbb{E}_\lambda \left[u_j(d_{jt}, q_{jt}^d, \Omega_{jt}) \right] + \beta \mathbb{E}_{\Omega_{jt+1}} \left[\bar{V}_{jt+1}(\Omega_{jt+1}) | \Omega_{jt}, d_{jt}, q_{jt} \right] + \varepsilon_{jt}^d$$

where $v_j^d(q_{jt}^d, \Omega_{jt})$ is the choice specific value function of choosing program d and quality score q_{jt}^d at time t . The first term is the expected flow utility under program d and quality score q_{jt}^d , as given by [Equation 10](#). The second term is the expected value of the future, discounted by $\beta \in (0, 1)$.

This depends on the ex-ante value function $\bar{V}_{jt+1}(\Omega_{t+1})$ which can be written as follows

$$\bar{V}_{jt+1}(\Omega_{jt+1}) = \mathbb{E}_\varepsilon \left[V_{jt+1}(\Omega_{jt+1}, \varepsilon_{jt+1}^d) \right] = \mathbb{E}_\varepsilon \left[\max_{d \in \{0,1,2\}} \left\{ \max_{q_{jt+1}^d \in (0,1]} v_j^d(q_{jt+1}^d, \Omega_{jt+1}) + \varepsilon_{jt+1}^d \right\} \right] \quad (15)$$

Using the arguments of [Hotz and Miller \(1989\)](#), [Arcidiacono and Miller \(2011\)](#) and [Arcidiacono and Ellickson \(2011\)](#), it can be shown that the expected value function can be written as difference between the choice-specific value function of an arbitrary choice h and the conditional choice probability of h

$$\bar{V}_{jt+1}(\Omega_{jt}) = v_j^h(q_{jt}^d, \Omega_{jt}) - \ln \left(P(d_{jt} = h | \Omega_{jt}) \right) + \gamma \quad (16)$$

where $v_j^h(q_{jt}^d, \Omega_{jt})$ is the choice-specific value function of option h , $P(d_{jt} = h | \Omega_{jt})$ is the probability that ACO j chooses option h conditional on Ω_{jt} (also referred to as Conditional Choice Probability (CCP)), and γ is the Euler constant. Substituting this expression for $\bar{V}_j(\Omega_{it+1})$ in $v_j^d(q_{jt}^d, \Omega_{jt})$ we can write the conditional value function of choice d as the sum flow utility, choice-specific value function of an arbitrary choice h and the conditional choice probability of h

$$v_j^d(q_{jt}^d, \Omega_{jt}) = \mathbb{E}_\lambda \left[u_j(d_{jt}, q_{jt}^d, \Omega_{jt}) \right] + \beta \mathbb{E}_{\Omega_{it+1}} \left[v_j^h(q_{jt}^d, \Omega_{jt}) - \ln \left(P(d_{jt} = h | \Omega_{jt}) \right) \middle| \Omega_{jt}, d_{jt}, q_{jt}^d \right] + \beta \gamma$$

Consider the continuation value function of an ACO that participates in the MSSP in year t . Since after exiting the MSSP the ACO cannot reenter, the choice of leaving the MSSP program can be modeled as a terminal action. The choice-specific value function of the terminal action h can be normalized to zero. Hence, the choice-specific value function of MSSP ($d_{jt} \in \{1, 2\}$) can be written using FFS ($d_{jt+1} = 0$) as the terminal action.¹⁰

The conditional choice probability of leaving the MSSP $P(d_{it} = 0 | \Omega_{it})$ can be seen as the empirical hazard rate of leaving the MSSP conditional of the state variables Ω_{it} . The transition and conditional choice probabilities can be empirically estimated using data on ACO's exit choice, as well as benchmark and risk score evolution over time.

The state space Ω_{it} contains 5 continuous variables: the historical benchmark b_{jt}^h , the rolling average spending $\tilde{y}_{jt}$, the Medicare regional spending y_{tj}^R and the update factors λ_{jt} . Let Ω_{jt}^c denote the subset of continuous state variables. To compute the expected value function using the terminal

¹⁰If in year t the ACO is still under FFS it takes a sequence of two actions to get to make the problem no longer dynamic. In particular, the sequence of actions $\{d_{jt+1} = 1, d_{jt+2} = 0\}$ is terminal. Hence, the expected value function under FFS can also be written in terms of the choice-specific value function and CCP of the terminal action. See Appendix D for details.

action approach, I need to discretize the variable in Ω_{jt}^c (Aguirregabiria and Mira, 2010).

I discretize each continuous state variable with a grid of K points. Then, the integrated choice-specific value function is given by

$$v_j^d(q_{jt}^d, \Omega_{jt}) = \mathbb{E}_\lambda \left[u_j(d_{jt}, q_{jt}^d, \Omega_{jt}) \right] + \beta \sum_{k=1}^K \left[F(\Omega_{jt}^c(k)) - F(\Omega_{jt}^c(k-1)) \right] \bar{V}(\Omega_{jt}(k)) \quad (17)$$

where $F(\Omega_{jt}^c(k))$ is the joint cdf of the variables in Ω_{jt}^c evaluated at the k -th point of the grid.

A marginal change in the quality score affects spending and in turn the rolling average spending. Therefore, the FOC can be derived as follows

$$\frac{\partial \mathbb{E}_\lambda u_j^d}{\partial q_{jt}}(d_{jt}, q_{jt}^d, \Omega_{jt}) + \beta \sum_{k=1}^K \left[f(\Omega_{jt+1}^c(k)) - f(\Omega_{jt+1}^c(k-1)) \right] \frac{\partial y_{jt}}{\partial q_{jt+1}} \bar{V}(\Omega_{jt}(k)) = 0 \quad (18)$$

where

$$\begin{aligned} f(\Omega_{jt}^c(k)) &= \frac{\partial F}{\partial y_{jt}}(\Omega_{jt}^c(k)) \\ \frac{\partial y_{jt}}{\partial q_{jt}} &= -\alpha \frac{y_{jt}}{q_{jt}} \\ \bar{V}(\Omega_{jt}(k)) &= -\ln \left(P(d_{jt} = 0 | \Omega_{jt}(k)) \right) + \gamma \end{aligned}$$

Distributional assumptions Since that value of the risk score and trend factors used to update the historical benchmark is revealed to the ACO only at the end of the performance year, I assume that the information set to predict the evolution of the adjustment factor possessed by the researcher is the same as that of the ACO. In particular, I assume that $\lambda_{jt} | \lambda_{jt-1} \sim N(\mu_{jt}^\lambda, \sigma_{jt}^\lambda)$ where μ_{jt}^λ and σ_{jt}^λ can be estimated with a linear regression of λ_{jt} on λ_{jt-1} and other patients' observable characteristics.

As described in Section 3, ACOs observe the realization of the reference spending choose the quality score to reduce spending. I model the reference spending y_{jt}^r as follows

$$\ln y_{jt}^r \sim N(\mu_{jt}, \sigma_{jt}^2) \quad \text{and} \quad \mu_{jt} = X_{jt} \beta_y$$

where X_{jt} is a vector of ACO characteristics, patients characteristics and ACO's past expenditure.

Taking log on both sides of Equation 6 and the distributional assumption on $\ln y_{jt}^r$ we obtain

the regression equation

$$\ln y_{jt} = X_{jt}\beta_y - \alpha \ln q_{jt} + \varepsilon_{jt}^y \quad (19)$$

which can be used to estimate α and β_y . Then I back out the estimated reference spending as

$$\bar{y}_{jt}^r = \hat{\beta}_y X_{jt}$$

The cost of care quality $C(q)$ varies across ACOs because the cost of care quality depends on the characteristics of the ACO and the ACO's patient population. Therefore, I assume that the cost parameter c is ACO-specific and that it can be modeled as a function of ACOs' observable characteristics $\mathbf{z}_{jt}$ and an additive error term ϵ_{jt}^c independent from $\mathbf{z}_{jt}$

$$\ln c_{jt} = \ln(\mathbf{z}_{jt})\beta_c + \delta_j + \epsilon_{jt}^c$$

where δ_j are ACO-level fixed effects and β_c are parameters to be estimated.

Finally, m and η are also parameters to be estimated.

Estimation Steps The estimation approach is an application of [Arcidiacono and Miller \(2011\)](#).

Then I recover the cost of quality parameter, the fixed cost of entry and the other utility parameters following these steps:

- (i) Estimate conditional choice probabilities (CCP) of the terminal action as the empirical hazard rate of leaving the MSSP. Estimate the transition probabilities with a linear regression

$$\lambda_{jt} = \beta_\lambda \lambda_{jt-1} + \mathbf{z}_{jt}\beta_z$$

- (ii) Estimate α and y_{jt}^r with the regression ([Equation 19](#)).

- (iii) Substitute the estimated CCP of exit in the continuation value function $\bar{V}(\Omega_{jt})$, and use the FOC as given by [Equation 18](#) to estimate the parameters of the utility function $\theta = (c, m, \eta)$. The details of the estimation methods are in Appendix D.

- (iv) Compute the choice specific value function of entry net of fixed costs

$$\tilde{v}^1(q_{jt}, \Omega_{jt}) = v^1(q_{jt}, \Omega_{jt}) - K(\mathbf{z}_j) \quad j \in \{1, 2\}$$

where $v^1(q_{jt}, \Omega_{jt})$ for $j \in \{1, 2\}$ are given by [Equation 17](#), and K_j is assumed to be a linear function of the ACO's observables $\mathbf{z}_j$, $K_j = \mathbf{z}_j\beta_K$. After normalizing the choice specific

value function of leaving the MSSP to zero, I can substitute the difference in the choice specific value functions ($v^1(q_{jt}, \Omega_{jt}) - v^0(q_{jt}, \Omega_{jt})$) in the logit probabilities of entry and estimate $K(\mathbf{z}_j)$ with maximum likelihood.

5.1 Estimation Results

Table 4 presents the estimated parameters of the regression equation

$$\ln c_{jt} = \ln(\mathbf{z}_{jt})\beta_c + \delta_j + \epsilon_{jt}^c$$

where δ_j are ACO-level fixed effects to control for unobserved heterogeneity. Column (1) and Column (2) show the result without FE, respectively. I observe that ACO-level FE explains a large part of the variation in the cost parameter. The average cost of exerting the maximum quality score ($q_{jt} = 1$) for a physician-led ACO with average values of benchmark, risk score, trend factors, and number of beneficiaries and no electronic health record system is about (192\$) per patient.

ACOs with higher spending have a higher cost of quality score. A negative change in patient health status increases the cost of quality score: a one percent increase in the risk score leads to a 3.46% increase in the cost parameter. ACOs led by Hospital and physician groups (mixed ACOs) and Hospital-led ACOs (hospital ACOs) have a cost of quality score that is 1.79% and 0.978% higher than that of ACOs led by physician groups only. ACOs that include providers that are part of a managed care organization have slightly higher cost of care quality. Moreover, ACOs equipped with an electronic health records system have lower cost of spending reduction by 1.36%.

Figure 6 illustrates how ACOs' selection under MSSP depends on the cost parameter c and the deviation between the Medicare regional spending and the ACO's FFS spending. Each dot represents the median value of the ratio between the FFS spending and the average Medicare regional spending y^{FFS}/y^R , for each ventile (5% bin) of the distribution of the cost parameter c . ACOs with values of

$$y^{FFS}/y^R$$

lower than one are likely to obtain a positive regional adjustment. We observe that the media value y^{FFS}/y^R is significantly lower among the ACOs that select into MSSP compared to the ACOs that stay under FFS. This shows further evidence of adverse selection induced by benchmark regionalization.

Entry costs are estimated with the latent variable model

$$\text{entry}_{jt} = 1 \iff \tilde{v}_1(q_{jt}, \Omega_{jt}) + FC_j + \varepsilon_{jt}^1 > v_0(q_{jt}, \Omega_{jt}) + \varepsilon_{jt}^0 \quad (20)$$

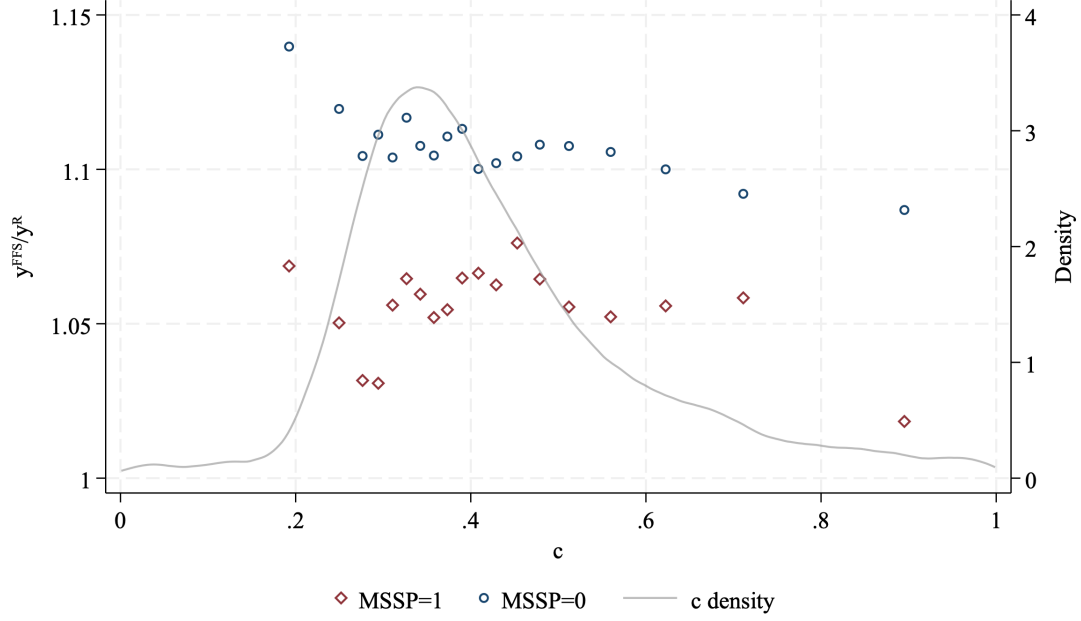


Figure 6: Selection under MSSP

Notes: The cost of quality score c is plotted on the x-axis and its empirical density is plotted on the right hand side. The regional deviation (measured as the ratio between the FFS spending and the Medicare average regional spending) y^{FFS}/y^R is plotted on the y-axis. (2) This figure shows the binned scatterplot of y^{FFS}/y^R and c . Each dots represents the median value of y^{FFS}/y^R for each ventile (5% bin) of the c distribution. The red-diamond shaped dots correspond to the value of the ACOs that select in, while the blue circle shaped dots correspond to the value of the ACOs that select out.

where FC_j is assumed to be a linear function of the ACO's observables $\mathbf{z}$, $FC_j = \mathbf{z}\beta_F$.

Since the value of $\tilde{v}_1(q_{jt}, \Omega_{jt})$ and $v_0(q_{jt}, \Omega_{jt})$ is estimated in the steps (i) - (iii), this reduces to estimating a logit model with $\tilde{v}_1(q_{jt}, \Omega_{jt}) - v_0(q_{jt}, \Omega_{jt})$ as an offset term.

Table 5 shows the estimated values of the β_F . I find that the ACOs with higher number of beneficiaries have higher entry costs. Mixed ACOs and Hospital ACOs have higher fixed costs than Physician-led ACOs. ACOs that have electronic health records system in place before joining the MSSP have lower entry costs. Moreover, ACOs that include hospitals with higher readmission rate have higher entry costs.

6 Counterfactuals

In this Section, I use the estimated model to perform a set of counterfactual exercises. First, I discuss how benchmark rebasement and regionalization affect participation into MSSP and Medicare savings. Secondly, I present and discuss the result of an alternative benchmarking policy that is designed to mitigate the ratchet effect while also preventing adverse selection. Finally, I evaluate the performance of the MSSP and the alternative policy compared to the counterfactual scenario where the MSSP does not exist, and all providers are under the standard FFS system.

6.1 Adverse Selection and Ratchet Effect

I use the results of these counterfactual simulations to evaluate the impact of adverse selection induced by benchmark regionalization and ratchet effect due to benchmark rebasement. I consider the “Simulated MSSP” as the status quo, and I assess their impact in terms of the total Medicare per-capita spending which is defined as the sum of the FFS spending per-capita and the shared savings payments per-capita. Let Y_j^M and Y_j^C denote the average total spending per-capita under the status quo and under the counterfactual scenario, respectively. Y_j^M is given by the sum of the per-capita spending y_j^M and the shared savings payments per-capita ss_j^M

$$Y_j^M = y_j^M + ss_j^M$$

Similarly, Y_j^C is given by the sum of the per-capita spending y_j^C and the shared savings payments per-capita ss_j^C

$$Y_j^C = y_j^C + ss_j^C$$

The key metric of interest is the percentage change in the total Medicare per-capita spending between the counterfactual scenarios and the actual MSSP. The latter is denoted as ΔY_j and can be written as follows:

$$\underbrace{\frac{Y_j^C - Y_j^M}{Y_j^M}}_{\Delta Y_j} = \underbrace{\frac{y_j^C - y_j^M}{Y_j^M}}_{\Delta y_j} + \underbrace{\frac{ss_j^C - ss_j^M}{Y_j^M}}_{\Delta ss_j} \quad (21)$$

where the first term is change in per-capita spending relative to the total Medicare spending (Δy_j), and the second term is the change in shared savings payments as a share of the total Medicare spending (Δss_j). Furthermore, Δy_j can be further decomposed as the sum of the change in spending of the ACOs that select in the MSSP Δy_j^I and that of the ACOs that select out of the MSSP Δy_j^O .

Results of the counterfactual simulations are reported in [Table 1](#). I weigh the ACO per-capita spending y_j^C and shared savings per-capita ss_j^C by the number of ACO's assigned beneficiaries so that the statistics in [Table 1](#) represent the overall impact of MSSP on Medicare spending. In both Panel A and B, columns one to six show the following variables: (1) P is the yearly average number of active ACOs from 2013 to 2022, (2) ΔY^I is the percentage change in spending per-capita among the ACOs that select in the MSSP (excluding the shared savings payments), (3) ΔY^O is the percentage change in spending per-capita among the ACOs that select out, (4) ΔY corresponds to the first term on the right hand side of [Equation 21](#), (5) ΔSS corresponds to the second term on the right hand side of [Equation 21](#), and (6) ΔNS is the percentage change in net spending per-capita as given by [Equation 21](#).

Ratchet Effect Benchmark rebasement influences both spending and participation. As discussed in Section 3, benchmark rebasement induces ACOs to delay spending reductions to prevent the benchmark from decreasing over time. Participation is also affected because ACOs have higher incentives to exit the MSSP as the benchmark declines over time. To estimate the impact of benchmark rebasement on net Medicare savings, I consider a counterfactual scenario where the benchmark is not rebased. However, the regional adjustment factor is still computed using the three-year rolling average spending $\tilde{y}_{jt}$. Specifically, the regionalized benchmark at the start of the second agreement period (AP) is given by

$$b_{jt+1}^r = (1 - w)b_{jt}^h + w(y_{tj}^R - \tilde{y}_{jt})$$

where b_{jt}^h is the (updated) historical benchmark from the first AP. Since $\tilde{y}_{jt}$ enters negatively in the calculation of b_{jt+1}^r , achieving a greater spending reduction during the first AP enables ACOs to secure a more positive regional adjustment. This, in turn, results in a higher benchmark for the second AP, mitigating the incentive to delay spending reductions.

[Figure 7](#) is analogous to [Figure 6](#) in Section 5 and it illustrates how selection depends on the cost of quality score and the ration y^{FFS}/y^R . Under this counterfactual scenario, selection patterns are similar to those observed in the actual MSSP. This is not surprising since the adverse selection through the regional adjustment is still in place. Participation, however, increases significantly from 41.2% to 50.5%, driven both by the entry of more ACOs with negative regional deviations and a lower exit rate. This can be attributed to the absence of rebasement, which enables ACOs to maintain higher benchmarks in the second AP for any given spending reduction achieved in the first AP.

Table 1: COUNTERFACTUALS
(Percentage change relative to status quo)

	Participation Rate in MSSP P	% Δ Spending Selected In Δy^I	% Δ Spending Selected Out Δy^O	% Δ Spending Gross SS Δy	Shared Savings Payments Δ_{SS}	% Δ Spending Net of SS ΔY
<i>Panel A: Voluntary MSSP</i>						
Simulated MSSP	41.2	0	0	0	0	0
No Rebasement	50.5	-0.91	-0.11	-1.02	0.12	-0.90
Naive Rebasement	36.8	-1.36	0.07	-1.29	0.18	-1.11
No Regionalization	39.2	-0.19	0.06	-0.13	-0.31	-0.44
Conditional Regionalization	40.7	-1.96	0.13	-1.83	-0.19	-2.02
<i>Panel B: Mandatory MSSP</i>						
Simulated MSSP	100	-0.76	-	-0.97	0.13	-0.85
No Rebasement	100	-1.11	-	-1.32	0.15	-1.17
Naive Rebasement	100	-1.28	-	-1.56	0.21	-1.35
No Regionalization	100	-0.43	-	-0.62	-0.14	-0.86
Conditional Regionalization	100	-2.16	-	-2.39	-0.07	-2.46

Notes: This table shows the percentage change in spending and shared savings payments between counterfactual scenarios and the simulated MSSP. I weight the ACO-level spending data by the number of assigned beneficiaries, so that the statistics are representative of the average value across ACOs. Panel A shows the results under the voluntary MSSP participation, while Panel B shows the results under the mandatory MSSP participation. In both Panel A and B, columns one to four shows the following variables: (1) P represents the yearly average number of active ACOs from 2013 to 2022, (2) ΔY^I is the average percentage change in spending per-capita among of the ACOs that select in the MSSP excluding the Shared Savings Payments, (3) ΔY^O is the average percentage change in spending per-capita among of the ACOs that select out, (4) Δy is the percentage change in spending per-capita among the all ACOs (5) ΔSS is the percentage change in Shared Savings Payments, and (6) ΔY is the percentage change in spending per-capita net of Shared Savings Payments among of the ACOs. In both Panel A and B, rows 1 report the results under the simulated MSSP regime with both benchmark rebasement and regionalization, and rows 2 to 5 report the results under the counterfactual four scenarios: (i) "No Rebasement" is the MSSP without the rebasement rule, (ii) in "Naive Rebasement" I consider MSSP with Benchmark Rebasement, but I impose that ACOs are not forward looking and maximize the static utility, (iii) "No Regionalization" is the MSSP without Benchmark Regionalization, and (iv) in "Conditional Regionalization" I impose that to qualify for a positive regional adjustment, ACOs need to have generated savings in the previous Agreement Period.

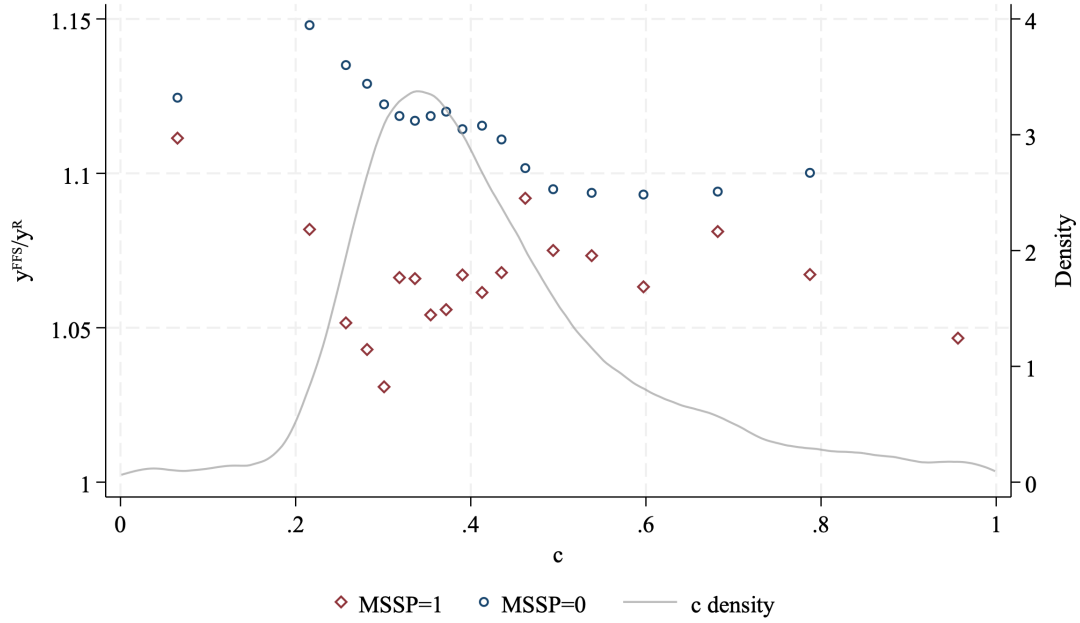


Figure 7: Selection under “No Rebasement”

Notes: The cost of quality score c is plotted on the x-axis, and its empirical density is plotted on the right-hand side. The regional deviation (measured as the ratio between the FFS spending and the Medicare average regional spending) y^{FFS}/y^R is plotted on the y-axis. (2) This figure shows the binned scatterplot of y^{FFS}/y^R and c . Each dot represents the median value of y^{FFS}/y^R for each ventile (5% bin) of the c distribution. The red diamond-shaped dots correspond to the value of the ACOs that select in, while the blue circle-shaped dots correspond to the value of the ACOs that select out.

The second row of Panel A in Table 7 reports the outcomes under this counterfactual scenario. Medicare net spending decreases by 0.90% relative to the status quo. The decomposition of ΔY shows that this is mainly due to the reduction in spending per-capita. As shown in the histogram in Figure 13, these savings result from spending reductions by ACOs that participate in both scenarios, while new entrants achieve only modest reductions. Moreover, the spending reductions are larger in the first AP and then level off in subsequent AP, which is the opposite of what we see in the “Simulated MSSP”. Under mandatory participation (Panel B), we observe a similar pattern, and net Medicare spending per-capita would have only a modest decrease from -0.90% to -1.17% .

These results confirm the presence of ratchet effect. In terms of per-capita spending, the ratchet effect reduces net Medicare savings by 117\$, which amounts to approximately 7.02 billion dollars. However, removing benchmark rebasement is not an ideal solution since spending reductions level off after the first AP, and it does not address the issue of adverse selection.

Adverse Selection The benchmark regionalization has two effects on MSSP net spending reduction. First, there is a direct effect through the benchmark of the ACOs that select in MSSP. ACOs whose historical spending is below the Medicare Regional average receive a positive regional adjustment. Hence, these ACOs can obtain shared savings payment by keeping their spending unchanged relative to FFS level. Second, there is an indirect effect through participation. ACOs whose spending is lower than the Medicare regional average are more likely to join and stay in the MSSP since they can more easily keep their spending below the benchmark. Similarly, ACOs whose spending is higher than the Medicare regional average are less likely to join the MSSP and more likely to exit.

Figure 8 illustrates selection under the counterfactual scenario where the benchmark is not rebased. We observe that for value of the cost parameter c above its mean, the ratio y^{FFS}/y^R is higher compared to the status quo. This reflects a shift in participation patterns: fewer ACOs with positive regional deviation join, while those with negative regional deviation are more likely to participate. These participation changes stem from different incentives. Without the regional adjustment factor, the ACO benchmark aligns more closely with counterfactual FFS spending, requiring ACOs to reduce spending below the FFS level to generate savings. As a result, adverse selection of “free shared savings” motivated ACO is less likely to occur in this counterfactual scenario. Furthermore, ACOs with spending above the regional average have greater incentive to participate, as their benchmarks are no longer penalized by a negative regional adjustment factor.

The fourth row of Panel A in Table 7 presents the MSSP savings in a counterfactual scenario where benchmark regionalization is not applied. Net Medicare spending decreases by 0.44% relative to the status quo, with approximately 70% of this reduction attributed to lower shared savings payments. This outcome arises because benchmarks are reduced when the positive regional adjustment is not applied. Since the ratchet effect is still present, expenditure among participating ACOs shows only a modest improvement compared to FFS (−0.23%). As illustrated in Figure 12 in Appendix F, additional spending reductions come primarily from new ACOs participating in the MSSP under this scenario. These ACOs tend to have spending above the regional average and have stronger incentive to participate since the negative regional adjustment is not applied. Conversely, ACOs that participate under both the status quo and the counterfactual scenario exhibit, on average, higher spending. This result can be explained by the fact that ratchet effect is even stronger in the absence of the regional adjustment.

These results indicate that adverse selection has negative impact on Medicare savings. Some ACOs participate in the MSSP to exploit regional adjustment and obtain shared savings payments without reducing their spending relative to the FFS level. In terms of spending per-capita, adverse

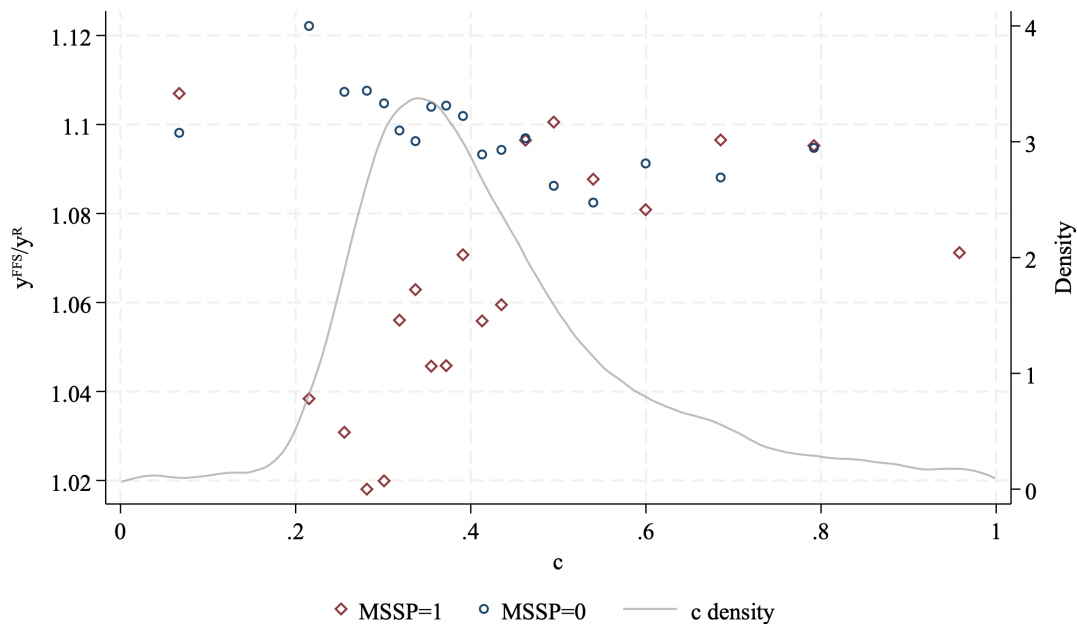


Figure 8: Selection under “No Regionalization”

Notes: The cost of quality score c is plotted on the x-axis, and its empirical density is plotted on the right hand side. The regional deviation (measured as the ratio between the FFS spending and the Medicare average regional spending) y^{FFS}/y^R is plotted on the y-axis. (2) This figure shows the binned scatterplot of y^{FFS}/y^R and c . Each dot represents the median value of y^{FFS}/y^R for each ventile (5% bin) of the c distribution. The red diamond-shaped dots correspond to the value of the ACOs that select in, while the blue circle-shaped dots correspond to the value of the ACOs that select out.

selection increases Medicare spending by around 57\$ per-capita, which corresponds to 3.43 billion dollars. Approximately 50% of that is the result of shared savings paid to ACOs that do not generate savings relative to the FFS level. Making the MSSP participation mandatory would not significantly improve Medicare savings as the additional ACOs that are forced to participate also have an incentive to delay spending reductions. As shown in Panel B, net Medicare savings would only have a modest increase in net Medicare savings compared to the voluntary scenario (from -0.44% to -0.62%).

6.2 Conditional Regionalization

The results of counterfactual simulations showed that tying ACO benchmarks to its past spending induces ACOs to delay spending reduction. Benchmark regionalization is not an ideal solution since it results in adverse selection, which costs Medicare additional payments to ACOs that do

not reduce spending relative to the FFS level. I propose an alternative benchmarking rule called Conditional Regionalization, which seeks to balance the trade-offs between rebasement and regionalization. This approach leverages regionalization to mitigate the ratchet effect while also including additional conditions to prevent adverse selection.

Let $\tilde{b}_{jt}^h$ denote the average benchmark of the first AP. An ACO is considered to have generated savings in a given AP if the three-year rolling average spending $\tilde{y}_{jt}$ is less than $\tilde{b}_{jt}^h$. Under “Conditional Regionalization” approach, the rebased benchmark at the start of the second AP is determined as follows:

$$b_{jt+1}^r = \begin{cases} (1-w)\tilde{y}_{jt} + w(y_{jt}^R - \tilde{y}_{jt}) & \text{if } \tilde{b}_{jt}^h - \tilde{y}_{jt} > 0 \quad \text{and} \quad y_{jt}^R - \tilde{y}_{jt} > 0 \\ b_{jt}^h & \text{if } \tilde{b}_{jt}^h - \tilde{y}_{jt} \leq 0 \quad \text{and} \quad y_{jt}^R - \tilde{y}_{jt} > 0 \\ \tilde{y}_{jt} & \text{if } \tilde{b}_{jt}^h - \tilde{y}_{jt} > 0 \quad \text{and} \quad y_{jt}^R - \tilde{y}_{jt} \leq 0 \\ (1-w)b_{jt}^h + w(y_{jt}^R - \tilde{y}_{jt}) & \text{if } \tilde{b}_{jt}^h - \tilde{y}_{jt} \leq 0 \quad \text{and} \quad y_{jt}^R - \tilde{y}_{jt} \leq 0 \end{cases}$$

where w is the weight assigned to the regional adjustment, y_{jt}^R is the regional average spending, and b_{jt}^h is the historical (updated) benchmark from the first AP.

The first two conditions are designed to eliminate the participation incentives for “free shared-savings”-motivated ACOs while still mitigating the ratchet effect. The regional adjustment factor is calculated using the standard MSSP approach. However, to address adverse selection, the MSSP rule is modified to require ACOs to achieve savings in the previous AP in order to qualify for a positive regional adjustment. As a result, ACOs with a positive regional deviation that generates savings in the first AP are subject to the standard benchmarking rule. In contrast, ACOs with a positive regional deviation that fail to generate savings are assigned the same historical (updated) benchmark as in the first AP.

Furthermore, the third and fourth conditions are intended to encourage the participation of ACOs whose historical spending exceeds the regional average. The negative regional adjustment is waived for ACOs that achieve savings in the prior AP, and their benchmark is fully rebased. However, the incentive to delay spending reductions is offset by the motivation to obtain a positive regional adjustment in the subsequent agreement period. If, instead, they fail to generate savings, then they receive a negative regional adjustment.

Figure 9 illustrates ACO selection under “Conditional Regionalization”. We observe that across the entire distribution of c , the ratio y^{FFS}/y^R is higher compared to the status quo. This reveals that participation increases among ACOs with negative regional deviation and decreases among those with positive regional deviation. Hence, the condition to qualify for a positive regional

adjustment is effective to mitigate the participation incentives of “free shared-savings” motivated ACOs. Moreover, this also shows that the negative regional adjustment discourages participation, leading many ACOs to exit the MSSP. Indeed, both under “No Regionalization” and “Conditional Regionalization” we observe a substantial increase in participation among ACOs with historical spending above the regional average.

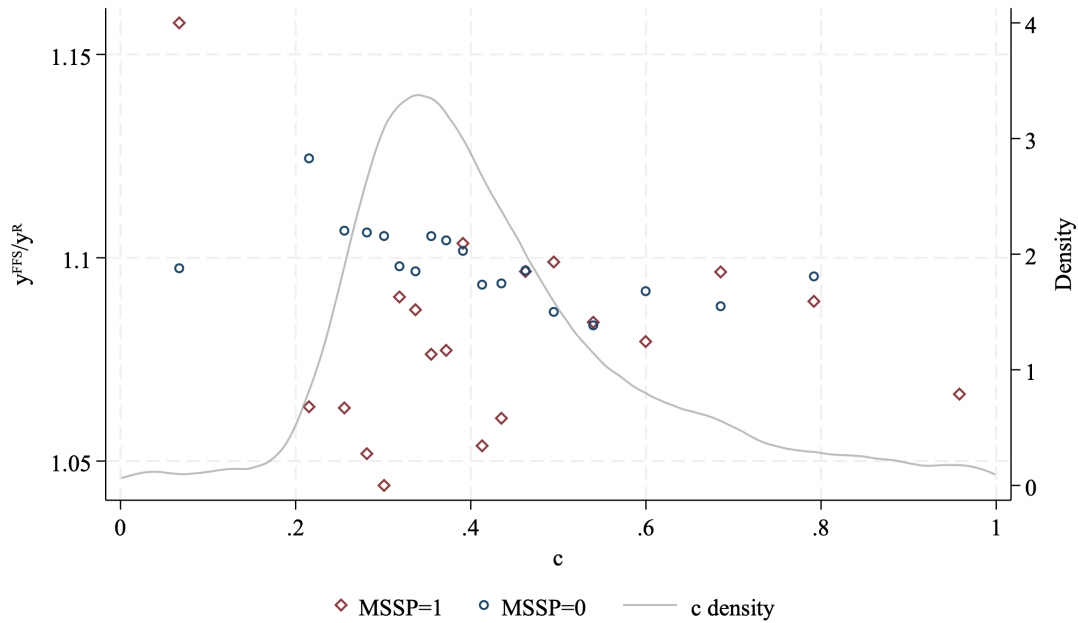


Figure 9: Selection under “Conditional Regionalization”

Notes: (1) The cost of quality score c is plotted on the x-axis, and its empirical density is plotted on the right hand side. The regional deviation (measured as the ratio between the FFS spending and the Medicare average regional spending) y^{FFS}/y^R is plotted on the y-axis. (2) This figure shows the binned scatterplot of y^{FFS}/y^R and c . Each dot represents the median value of y^{FFS}/y^R for each ventile (5% bin) of the c distribution. The red diamond-shaped dots correspond to the value of the ACOs that select in, while the blue circle-shaped dots correspond to the value of the ACOs that select out.

The fifth row of Panel A in Table 7 summarizes the counterfactual results under the proposed benchmarking rule. Net Medicare spending decreases by 2.02% relative to the status quo. The decomposition of ΔY reveals that despite the alternative policy reduces Medicare spending per-capita by an additional 1.83%, shared savings payments per-capita ΔSS decrease by 0.19% relative to the status quo. The alternative policy generates additional net Medicare savings of 260\$ per-capita compared to the status quo, amounting to approximately 15.6 billion dollars. As shown in Panel B, under mandatory participation, net Medicare savings relative to the status quo only increase by an additional 23% compared to voluntary participation. This indicates the ACOs that self-select in

the MSSP are able to generate almost the maximum possible savings under the proposed rule.

Figure 15 in Appendix F shows that this additional spending reduction is driven by both existing ACOs and new entrants participating in the MSSP under this counterfactual scenario. Unlike the actual MSSP, spending reduction continues over time, although at a decreasing rate. This trend suggests that the additional conditions for applying the positive regional adjustment effectively mitigate the ratchet effect. Furthermore, changes in ACO selection and reductions in shared savings payments compared to the status quo demonstrate that these conditions also help prevent adverse selection.

MSSP versus FFS I conclude this section by evaluating the outcome of the MSSP and the “Conditional Regionalization” relative the counterfactual scenario where the MSSP does not exist, and all providers are under FFS. The statistics presented in Table 7 are relative to the counterfactual Medicare spending under mandatory FFS.

Under voluntary participation (Panel A), Medicare spending per-capita under MSSP is approximately 1.16% lower compared to FFS spending (8.26 billion dollars), and shared savings payments represent 0.13% of per-capita spending. Under mandatory participation (Panel B), Medicare spending decreases by an additional 25% compared to the voluntary scenario (11.5 billion dollars).

The alternative policy, “Conditional Regionalization”, outperforms the simulated MSSP and other counterfactual scenarios both in terms of Medicare spending reductions and shared savings payments. Under voluntary participation, Medicare spending per-capita decreases by 2.96% relative to FFS, resulting in 23.1 billion dollars in savings. Despite this more than doubling the reduction in spending compared to the “Simulated MSSP”, shared savings payments under “Conditional Regionalization” represent only 0.08% of per-capita spending, while they represent 0.13% of per-capita spending under the “Simulated MSSP”. Under mandatory participation, Medicare total savings relative to FFS increase by an additional 20% (26.8 billion dollars) compared to the voluntary participation scenario.

These results highlight that, despite challenges such as the ratchet effect and adverse selection, both the current MSSP and the alternative policy “Conditional Regionalization” could yield significant savings if participation incentives were strengthened. Future research should explore methods to incentivize participation under the current program design or alternative frameworks, as well as investigate the relationship between these incentives and entry costs.

7 Conclusions

Incentive payment programs and pay-for-performance contracts are becoming increasingly popular as a form of Government regulation to improve the efficiency and quality of healthcare delivery. The Medicare Shared Savings Program is a voluntary incentive payment contract designed to induce providers to form integrated healthcare organizations, called Accountable Care Organizations, to reduce unnecessary health services through improved care coordination among providers. Under this program, ACO providers are rewarded for keeping the average per-capita spending below a benchmark based on the ACO providers' spending under the fee-for-service system.

This study examines the impact of benchmarking rules on ACO participation and spending decisions under the Medicare Shared Savings Program (MSSP), and provides theoretical and empirical insights on how these rules influence program effectiveness and overall Medicare expenditures. Specifically, the findings emphasize two significant issues: the ratchet effect induced by benchmark rebasement and adverse selection stemming from benchmark regionalization.

The ratchet effect occurs because the MSSP's rebasement mechanism ties future benchmarks to past spending, incentivizing ACOs to moderate their initial cost-reduction efforts to secure more favorable benchmarks in subsequent periods. Simulations reveal that this dynamic distorts incentives, leading to delayed savings and reduced participation in the MSSP. Adjustments to rebasement rules, such as excluding ACOs' own spending from future benchmark calculations, could mitigate these inefficiencies, as shown by counterfactual scenarios.

Benchmark regionalization, introduced to address the ratchet effect, can result in adverse selection. Empirical evidence suggests that ACOs receiving positive regional adjustments are more likely to participate, while those with negative adjustments are more likely to exit the program. The resulting increase in shared savings payments for high-cost ACOs inflates Medicare spending without commensurate improvements in cost efficiency. The counterfactual analyses demonstrate that removing benchmark regionalization reduces per-capita Medicare spending, primarily through lower shared savings payments.

To balance these conflicting necessities to mitigate the ratchet effect and prevent adverse selection, I propose an adjustment to the current MSSP design called Conditional Regionalization. This alternative benchmarking rule applies regional adjustments only to ACOs that generate genuine cost savings. I estimate that this policy effectively addresses adverse selection while mitigating the ratchet effect, offering a pathway to enhance the MSSP's net savings potential.

The findings underscore the importance of carefully aligning incentives in value-based payment models to balance participation, cost containment, and care quality. While this study focuses on

the MSSP, its insights have broader relevance to other Medicare initiatives, such as the Prospective Payment System and the Bundled Payments for Care Improvement Initiative. Future research could build on this work by incorporating interactions among providers within an ACO or exploring the implications of similar incentive structures in other healthcare contexts.

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Appendix A

Historical Benchmark

In order to compute the HB, the expenditure in the first two benchmark years (BY1 and BY2) are trended (using the national growth rate) and risk adjusted (using the HCC risk score) to the last benchmark year (BY3). The weights for BY1, BY2 and BY3 are 0.1, 0.3, 0.6 respectively.

Let t be the entry year and y_t denote the ACO's spending in year t . Then the benchmark years are $t - 3$, $t - 2$, and $t - 1$. The historical benchmark for the first performance year is

$$hb_t = 0.1y_{t-3} + 0.3y_{t-2} + 0.6y_{t-1}$$

where y_{t-3} and y_{t-2} are trended and risk adjusted to $t - 1$.

At the start of each performance year, the historical benchmark is also adjusted to account for changes in the ACO's participants list and changes in the population of assigned beneficiaries ¹¹.

Rebased Historical Benchmark

At the end of AP1 the ACO has the option to enter a second Agreement Period (AP2) of 3 years. The benchmark in AP2 is recalculated based on the most recent 3 years of expenditure, i.e. the expenditure in the performance years of AP1.

Let t be the first year of the second AP. Then the Historical Rebased Benchmark is given by

$$hb_t = \frac{1}{3}y_{t-3} + \frac{1}{3}y_{t-2} + \frac{1}{3}y_{t-1}$$

where y_{t-3} and y_{t-2} are trended and risk adjusted to $t - 1$ as for the first AP ¹².

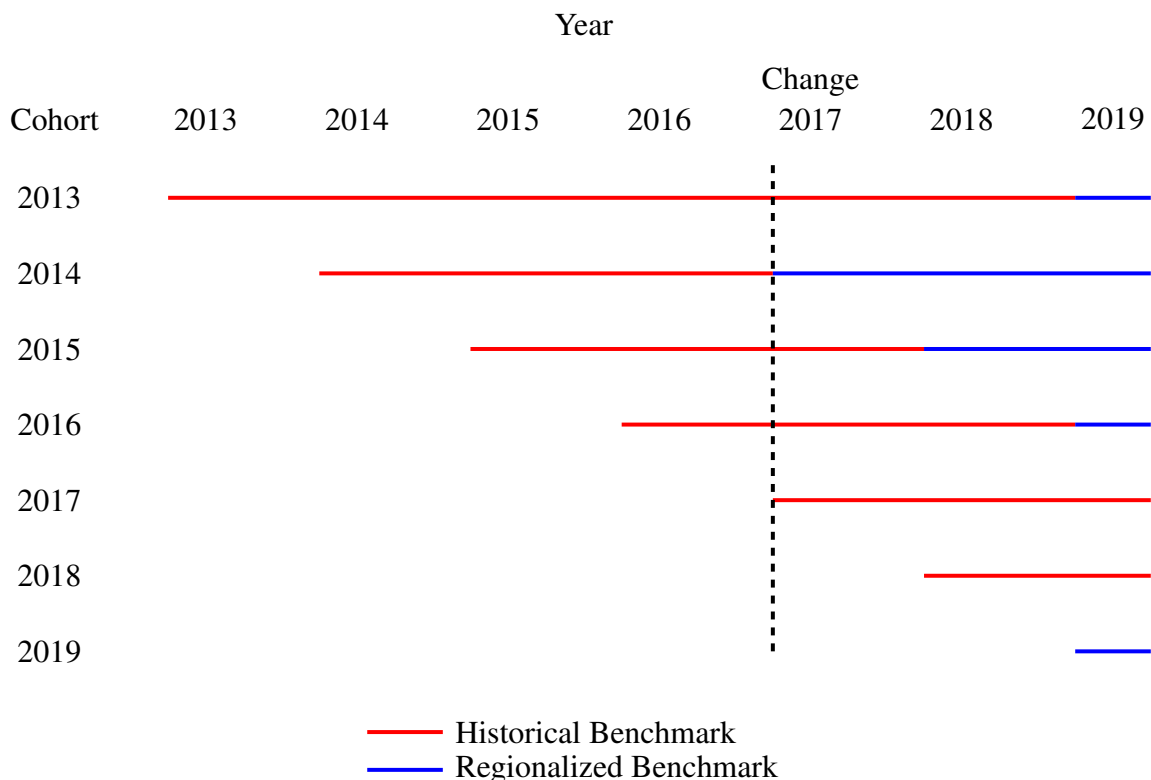
Regional Adjustment

The Regional Adjustment was introduced starting from 2017 and was not part of the original MSSP rules. Since the Regional Adjustment applies starting from the second Agreement Period, different cohort of ACOs are affected by this change at different times. For example, for the 2014 cohort the benchmark regionalization applies starting from 2017, while for the 2015 cohort the benchmark regionalization applies starting from 2018.

¹¹Clearly, there is no adjustment in the first performance year

¹²Notice the in the second AP uses equal weights for the benchmark years

There is one exception to this rule. For 2013 cohort the Benchmark Regionalization applies starting from the third Agreement Period (i.e. from 2019 onward).



Quality Measures

An ACO will earn quality points on a sliding scale based on level of performance. As shown in Table 2: performance below the minimum attainment level (the 30th percentile) for a measure will receive zero points for that measure; performance at or above the 90th percentile of the performance benchmark earns the maximum points available for the measure.

The total points earned for measures in each domain, including any quality improvement points, will be summed and divided by the total points available for that domain to produce an overall domain score of the percentage of points earned relative to points available. The percentage score for each domain will be averaged together to generate a final overall quality score for each ACO that will be used to determine the amount of savings it shares or, if applicable, the amount of losses it owes.

Domain	Number of Individual Measures	Total Measures for Scoring Purposes	Total Possible Points	Domain Weight
Patient/Caregiver Experience	8	8 individual survey module measures	16	25%
Care Coordination/ Patient Safety	10	10 measures, the EHR measure is double-weighted (4 points)	22	25%
Preventive Health	8	8 measures	16	25%
At-Risk Population	7	6 measures, including a 2-component diabetes composite measure	12	25%
Total in all Domains	33	32	66	100%

Figure 10: Quality Score Domains

Appendix B

ACOs Entry and Exit over time

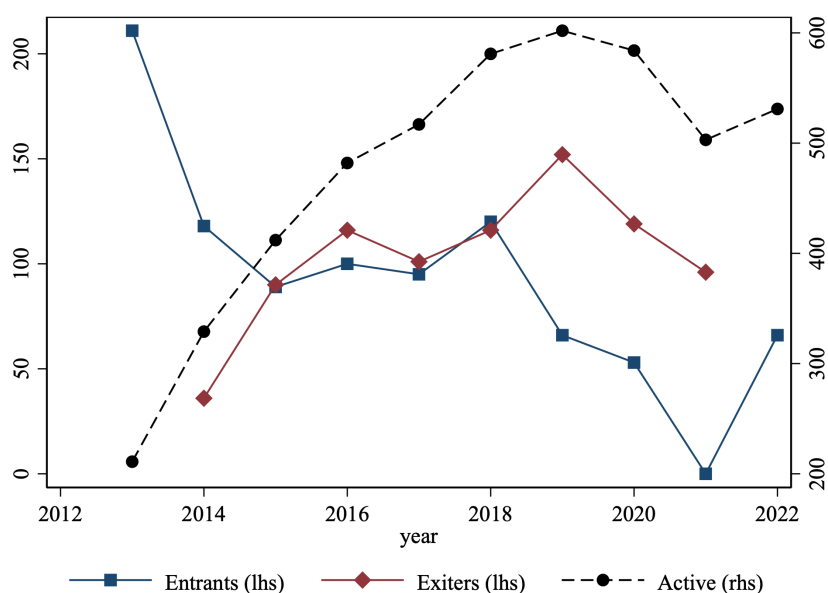


Figure 11: ACOs Entry and Exit

Notes: The blue and red line are plotted on the left-axis and show the number of ACOs that enter and exit respectively. The black dashed line is plotted on the right axis and shows the total number of active ACOs.

Table 2: ACOs Summary Statistics

	2013	2014	2015	2016	2017	2018	2019	Total
prov (1000)	4.421 (6.286)	3.933 (5.825)	4.647 (6.563)	4.350 (5.996)	7.166 (9.767)	7.696 (10.92)	8.486 (14.01)	6.136 (9.601)
benef (1000)	16.75 (15.35)	16.08 (15.16)	18.55 (18.51)	18.25 (17.83)	19.05 (19.12)	18.42 (18.57)	21.05 (22.17)	18.56 (18.66)
risk score	1.072 (0.0945)	1.071 (0.103)	1.069 (0.103)	1.075 (0.104)	1.002 (0.0876)	1.006 (0.0883)	1.013 (0.0942)	1.038 (0.101)
two-sided	0.0228	0.00909	0.00765	0.0509	0.0826	0.0785	0.0821	0.0537
bench (1000)	11.79 (3.086)	10.49 (2.597)	10.41 (2.483)	10.70 (3.326)	10.67 (2.230)	10.76 (2.171)	10.80 (2.227)	10.74 (2.578)
sav rate	0.00396 (0.0560)	0.00589 (0.0493)	0.00647 (0.0586)	0.00898 (0.0541)	0.0135 (0.0480)	0.0247 (0.0322)	0.0247 (0.0465)	0.0143 (0.0493)
sav above min	0.260 (0.440)	0.276 (0.448)	0.306 (0.461)	0.310 (0.463)	0.343 (0.475)	0.380 (0.486)	0.503 (0.501)	0.353 (0.478)
qual score	0.982 (0.0852)	0.915 (0.0795)	0.934 (0.0908)	0.946 (0.0737)	0.924 (0.0737)	0.929 (0.0688)	0.944 (0.0248)	0.936 (0.0734)
ACOs	219	330	392	432	472	548	475	2868
mean coefficients; sd in parentheses								

Table 3: Effect of quality on high cost services

	(1) Readm	(2) EDV hosp	(3) IP	(4) OP	(5) Ambul
quality	-0.018*** (0.001)	-0.025*** (0.003)	-0.019*** (0.002)	-0.035*** (0.004)	-0.031*** (0.005)
admissions	0.242*** (0.010)	1.081*** (0.024)	0.822*** (0.018)	-0.447*** (0.033)	0.427*** (0.041)
ED visits	-0.074*** (0.009)	-0.146*** (0.020)	-0.036** (0.015)	0.773*** (0.028)	0.904*** (0.034)
risk score	0.041** (0.017)	0.515*** (0.039)	-0.043 (0.029)	0.166*** (0.054)	0.202*** (0.067)
beneficiaries	0.003* (0.002)	0.021*** (0.004)	-0.000 (0.003)	0.021*** (0.006)	-0.010 (0.007)
mixed ACO	-0.013*** (0.003)	-0.073*** (0.007)	0.023*** (0.005)	0.223*** (0.009)	-0.009 (0.011)
hospital ACO	-0.016*** (0.004)	-0.039*** (0.010)	-0.011 (0.007)	0.151*** (0.014)	-0.032* (0.017)
Constant	1.834*** (0.041)	0.122 (0.096)	-0.994*** (0.070)	0.544*** (0.132)	-8.081*** (0.162)
Obs	4178	4188	4188	3977	4188
R-squared	0.28	0.66	0.63	0.34	0.48

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Selection on Levels

	Odds Ratio
pos reg adj	0.334*** (0.050)
sav above min	0.992*** (0.002)
risk ratio	1.069*** (0.020)
historical benchmark	0.876*** (0.044)
rolling avg spending	1.128** (0.059)
beneficiaries	0.758*** (0.064)
mixed ACO	0.843 (0.111)
hospital ACO	0.474*** (0.124)
AP 2	4.404*** (1.179)
AP 3	8.657*** (4.237)
AP 4	1.289 (1.520)
Obs	3834

Exponentiated coefficients; Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix C

Value Function and CCP

The state variables Ω_t and the choices d_t and q_t are observed by the researcher, while ϵ_t are only observed by the agent. The ACO is forward-looking and discounts future payoffs using the discount factor β . The agent has rational expectations about the evolution of b_t , $\bar{q}_t$, r_t and ϵ_t , and chooses the actions d_t and q_t in each period so as to maximize expected discounted future utility.

The timing of the model is such that the discrete choice d_t is made at the beginning of the period, prior to making the continuous choice q_t . Let q_t^d denote the optimal q_t given Ω_t and d_t .

The value function from the perspective of the beginning of the period can be expressed recursively using the ex-ante value function

$$\bar{V}(\Omega_t) = \mathbb{E} \left[V(\Omega_t, \epsilon_t) | x_t \right] = \max_{d_t \in \mathcal{D}} \{ v(d_t, q_t^d, \Omega_t) + \epsilon_t^d \}$$

where $v(d_t, q_t^d, \Omega_t)$ is the choice specific value function defined as

$$v(d_t, q_t^d, \Omega_t) = \max_{q_t} \left\{ u(d_t, q_t, \Omega_t) + \beta \mathbb{E}_{\Omega_{t+1}} \left[\bar{V}(\Omega_{t+1}) | d_t, \Omega_t, q_t \right] \right\}$$

I assume the ϵ_t s follow a type I extreme value distribution, yielding the classic dynamic logit model. The probability of choice $d_t = d$ conditional on the state variables (which we will refer to as CCP) is then given by

$$\begin{aligned} p(d | \Omega_t) &= \frac{\exp v(d, q_t^d, \Omega_t)}{\sum_{d' \in \mathcal{D}} \exp v(d', q_t^{d'}, \Omega_t)} \\ &= \frac{1}{\sum_{d' \in \mathcal{D}} \exp [v(d', q_t^{d'}, \Omega_t) - v(d, q_t^d, \Omega_t)]} \end{aligned}$$

and the ex-ante value function can be written as

$$\bar{V}(\Omega_t) = \ln \left(\sum_{d' \in \mathcal{D}} \exp v(d', q_t^{d'}, \Omega_t) \right) + \gamma$$

where γ is the Euler constant.

The ex-ante value function can be rewritten with respect to the conditional choice of an arbitrary

choice $d_t = h$

$$\begin{aligned}
\bar{V}(\Omega_t) &= \ln \left(\exp v(h, q_t^h, \Omega_t) \left(\frac{\sum_{d' \in \mathcal{D}} \exp v(d', q_t^{d'})}{\exp v(h, q_t^h, \Omega_t)} \right) \right) + \gamma \\
&= \ln \left(\sum_{d' \in \mathcal{D}} \exp \left[v(d', q_t^{d'}, \Omega_t) - v(h, q_t^h, \Omega_t) \right] \right) + v(h, q_t^h, \Omega_t) + \gamma \\
&= -\ln \left[p(h|x_t) \right] + v(h, q_t^h, \Omega_t) + \gamma
\end{aligned}$$

Substituting for $\bar{V}(\Omega_{t+1})$ in the the choice specific value function we obtain

$$v(d, q_t^d, \Omega_t) = u(d, q_t^d, \Omega_t) + \beta \int \left(v(h, q_{t+1}^h, \Omega_{t+1}) - \ln \left[p(h|\Omega_{t+1}) \right] \right) f(\Omega_{t+1}|\Omega_t, q_t^d) d\Omega_{t+1}$$

where $d_{t+1} = h$ is an arbitrary choice in $t + 1$. The future value term now has three components: the function characterizing the transitions of the state variables, the CCPs for the arbitrary choice, and the conditional value function associated with the arbitrary choice.

Using the arguments of [Hotz and Miller \(1989\)](#), [Arcidiacono and Miller \(2011\)](#) and [Arcidiacono and Ellickson \(2011\)](#) it can be shown that dealing with the remaining conditional value function is surprisingly straightforward. The key to the argument is that the researcher can choose to which choice (and hence which conditional value function) to make the future value term relative. Because discrete choice models are estimated using differences in conditional value functions, in many cases a clever choice of conditional value function for this benchmark can allow the difference in the future utility terms across two choices to be characterized by the one period ahead CCPs alone.

What we need is that the choice set includes an action (or a sequence of actions) that, when taken, implies that no further decisions are made. In other words, once this terminal action is chosen, the agent's decision problem is no longer dynamic, allowing the future value term to be replaced with a known parametric form or normalized to zero.

Case 1: After joining MSSP

When the ACO is participating in the MSSP the choice $d_t = 0$ is a terminal action because once the ACO exit from the MSSP it stays under FFS forever and the problem is no longer dynamic. Hence, we can write the choice specific value function for $d_t = 1$ can be written using $d_{t+1} = 0$ as the arbitrary choice.

$$v_1(\Omega_t) = u_1(q_{1t}, \Omega_t) + \beta \int \left(v_0(\Omega_{t+1}) - \ln [P_0(\Omega_{t+1})] \right) f(\Omega_{t+1} | \Omega_t, q_t^1) d\Omega_{t+1} + \beta\gamma$$

The conditional value function for choice $d_{t+1} = 0$ does not have a future value component because the exit decision terminates the dynamic decision process. Since the flow utility is normalized to zero $v_0(\Omega_{t+1})$ Hence,

$$v_1(\Omega_t) = u_1(q_{1t}, \Omega_t) - \beta \int \left(\ln [P_0(\Omega_{t+1})] \right) f(\Omega_{t+1} | \Omega_t, q_{1t}) d\Omega_{t+1} + \beta\gamma$$

Case 2: Before joining MSSP

When the ACO is still under FFS it takes a sequence of two actions to get to make the problem no longer dynamic. In particular the sequence of actions $\{d_{t+1} = 1, d_{t+2} = 0\}$ is terminal. Hence, we can write the choice specific value function for $d_t = 0$ can be written using $d_{t+1} = 1$ as the arbitrary choice for year $t + 1$

$$v_0(\Omega_t) = \beta \int \left(v_1(\Omega_{t+1}) - \ln [p(1 | \Omega_{t+1})] \right) f(\Omega_{t+1} | \Omega_t, d_t) d\Omega_{t+1} + \beta\gamma$$

and $d_{t+2} = 0$ as the arbitrary choice for year $t + 2$

$$\begin{aligned} v_0(\Omega_t) &= \beta \int \left(u_1(q_{1t+1}, \Omega_{t+1}) - \ln [P_1(\Omega_{t+1})] \right) f(\Omega_{t+1} | \Omega_t, d_t) d\Omega_{t+1} + \beta\gamma \\ &+ \beta^2 \int \int \left(v_0(\Omega_{t+2}) - \ln [P_0(\Omega_{t+2})] \right) f(\Omega_{t+2} | \Omega_{t+1}, d_{t+1}, q_{1t+1}) f(\Omega_{t+1} | \Omega_t, d_t) d\Omega_{t+2} d\Omega_{t+1} + \beta^2\gamma \end{aligned}$$

where, as above, $v_0(\Omega_{t+2}) = 0$. Then

$$\begin{aligned} v_0(\Omega_t) &= \beta \int \left(u_1(q_{1t+1}, \Omega_{t+1}) - \ln [P_1(\Omega_{t+1})] \right) f(\Omega_{t+1} | \Omega_t, d_t) d\Omega_{t+1} + \beta\gamma \\ &- \beta^2 \int \int \left(\ln [P_0(\Omega_{t+2})] \right) f_{\Omega_{t+2} | \Omega_{t+1}, d_{t+1}, q_{1t+1}} f(\Omega_{t+1} | d_t) d\Omega_{t+2} d\Omega_{t+1} + \beta^2\gamma \end{aligned}$$

Logit Probabilities

We can compute the difference in choice specific value functions

$$\begin{aligned}
v_1(\Omega_t) - v_0(\Omega_t) &= u_1(q_{1t}, \Omega_t) - \beta \int u_1(q_{1t+1}, \Omega_{t+1}) f(\Omega_{t+1} | \Omega_t, d_t) d\Omega_{t+1} \\
&+ \beta \int \left(\ln \left[\frac{P_1(\Omega_{t+1})}{P_0(\Omega_{t+1})} \right] \right) f(\Omega_{t+1} | \Omega_t, d_t) d\Omega_{t+1} \\
&+ \beta^2 \int \int \left(\ln \left[P_0(\Omega_{t+2}) \right] \right) f(\Omega_{t+2} | \Omega_{t+1}, d_{t+1}) f(\Omega_{t+1} | \Omega_t, d_t) d\Omega_{t+2} d\Omega_{t+1} - \beta^2 \gamma
\end{aligned}$$

which can be used as input in the logit probabilities

$$P_1(\Omega_t) = \Pr(d_t = 1 | \Omega_t) = \frac{1}{1 + \exp(v_1(\Omega_t) - v_0(\Omega_t))}$$

Appendix D

Estimation of the utility parameters

Given α , I treat the FOC on the choice specific value function $\frac{\partial v_M}{\partial q} = 0$ as a moment condition. I estimate β_c and (m, η) iteratively through this iterative procedure. I first initialize (m, η) and repeat the following steps

(i) compute c from the FOC

$$c = \frac{1}{q^2} \left[-m\alpha y_M + \eta(1 - q^{-\alpha})\alpha y_M^2 + q \frac{\partial SS}{\partial q} - \alpha y_M \left(\frac{\partial SS}{\partial y} + \frac{\beta}{N} \frac{\partial EV}{\partial y} \right) \right]$$

(ii) estimate β_c by regressing c on a set of ACO observables Z and compute $\hat{c} = g(Z\hat{\beta}_c)$

(iii) I replace the estimated $\hat{c}$ in the FOC and estimate (m, η) using GMM.

I repeat this procedure until the change of the estimated parameters across successive iterations is below some threshold.

Algorithm 1 Estimate (c, m, η)

initialize $(\hat{m}, \hat{\eta})$ and Δ ;

while $\Delta > \epsilon$ **do**

 Given $\hat{m}, \hat{\eta}$, compute c from FOC

 Estimate β and compute $\hat{c} = g(Z\hat{\beta}_c)$;

 Replace $\hat{c}$ in FOC and estimate (m, η) using GMM;

 Compute $\Delta = \max$ change in $(\hat{\beta}_c, \hat{m}, \hat{\eta})$ from previous iteration

end

Appendix F

Table 5: Cost of Quality Parameter

	(1)	(2)
benchmark	4.600*** (0.205)	2.029*** (0.114)
risk ratio	3.465*** (0.598)	3.489*** (0.567)
trend fact	4.885*** (0.281)	5.563*** (0.276)
beneficiaries	0.110** (0.046)	0.180*** (0.023)
mixed ACO	1.790*** (0.362)	1.168*** (0.332)
hospital ACO	0.978*** (0.256)	0.622*** (0.221)
mngd care	0.618*** (0.178)	0.500*** (0.149)
ehr	-1.366*** (0.273)	-1.192*** (0.256)
alt pay	0.063 (0.057)	0.062 (0.048)
Constant	-12.205*** (0.480)	-5.978*** (0.257)
Obs	5001	5001
R-squared	0.597	0.136
FE	Yes	No

I control for unobserved heterogeneity using ACO level FE

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Entry Costs

	(1) entry
size	0.147*** (0.035)
mixed ACO	3.142*** (0.221)
hospital ACO	1.774*** (0.300)
alt pay	0.071 (0.138)
mngd care	1.314*** (0.201)
ehr	-2.630*** (0.215)
readm level 2	0.162 (0.120)
readm level 3	0.256** (0.128)
readm level 4	0.539*** (0.138)
Constant	0.494*** (0.108)
Observations	5408

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix F

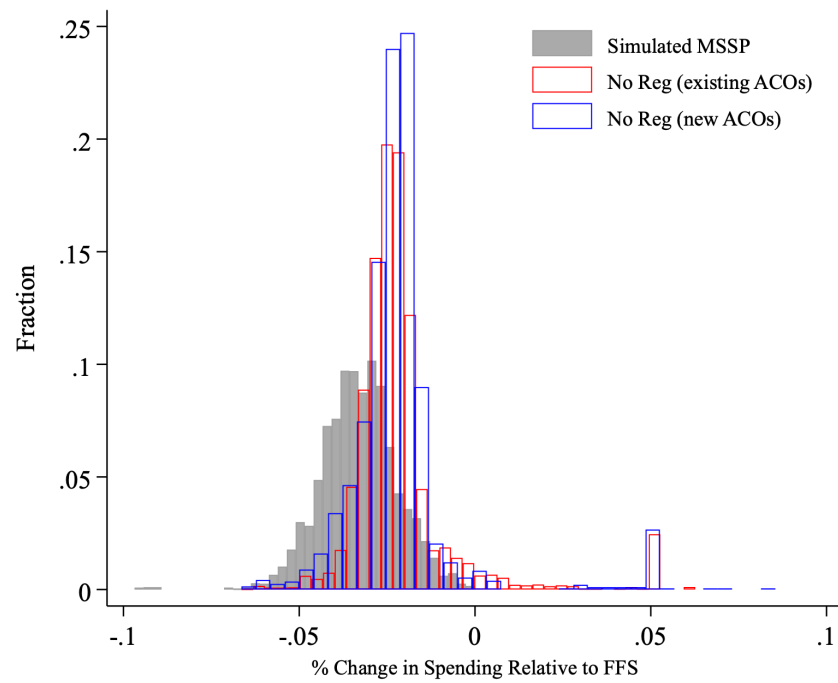


Figure 12: No Regionalization vs. Simulated MSSP

Notes: (1) Simulated MSSP and No Regionalization histograms include ACOs that participate in both actual MSSP and the counterfactual scenario.
(2) No Regionalization (new ACOs) histogram include ACOs that participate in the counterfactual scenario but not in the actual MSSP.

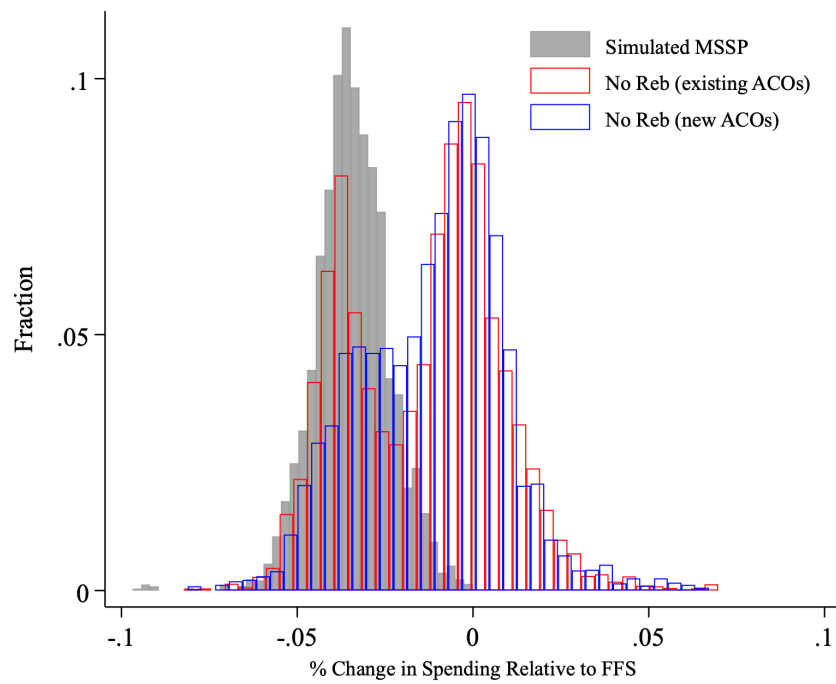


Figure 13: No Rebasement vs. Simulated MSSP

Notes: (1) Simulated MSSP and No Rebasement histograms include ACOs that participate in both actual MSSP and the counterfactual scenario.
 (2) No Rebasement (new ACOs) histogram include ACOs that participate in the counterfactual scenario but not in the actual MSSP.

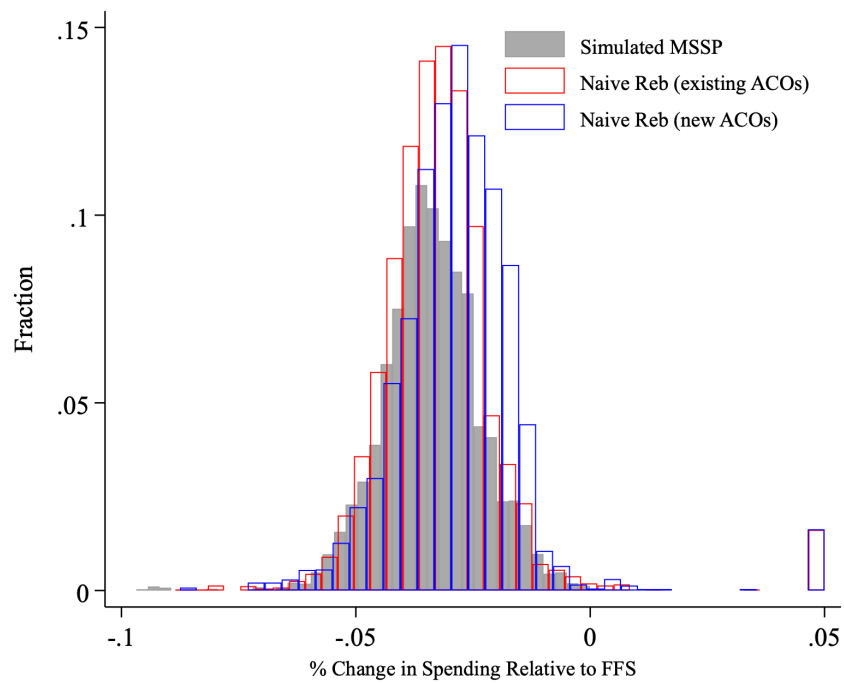


Figure 14: Naive Rebasement vs. Simulated MSSP

Notes: (1) Simulated MSSP and Naive Rebasement histograms include ACOs that participate in both actual MSSP and the counterfactual scenario.
 (2) Naive Rebasement (new ACOs) histogram include ACOs that participate in the counterfactual scenario but not in the actual MSSP.

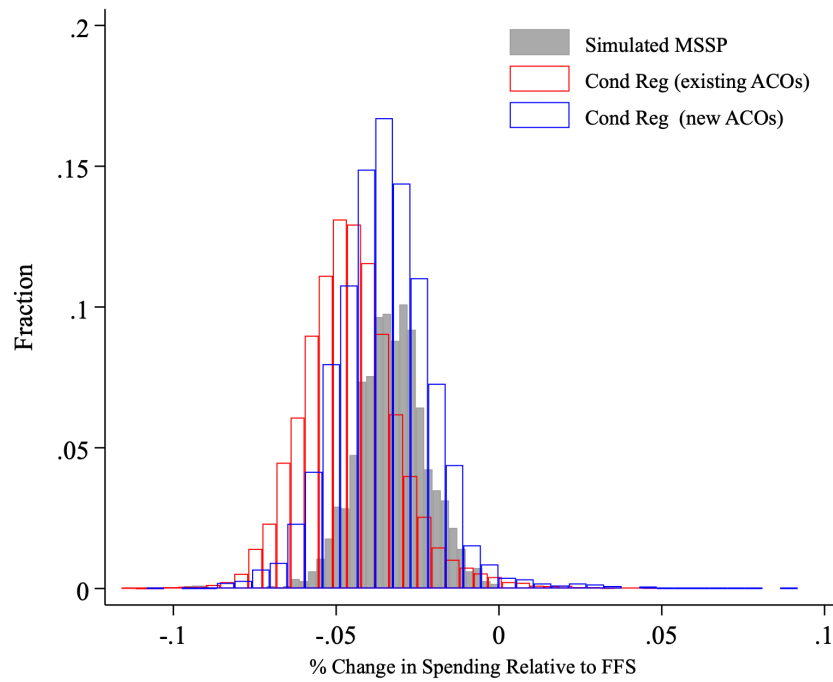


Figure 15: Conditional Regionalization vs. Simulated MSSP

Notes: (1) Simulated MSSP and Conditional Regionalization histograms include ACOs that participate in both actual MSSP and the counterfactual scenario. (2) Conditional Regionalization (new ACOs) histogram include ACOs that participate in the counterfactual scenario but not in the actual MSSP.

Table 7: COUNTERFACTUALS
(Percentage change relative to FFS)

	Participation Rate in MSSP P	% Δ Spending Gross SS Δy	% Δ Shared Savings Payments ΔSS	% Δ Spending Net of SS ΔY
<i>Panel A: Voluntary MSSP</i>				
Simulated MSSP	41.2	-1.16	0.13	-1.06
No Rebasement	50.5	-2.05	0.16	-1.89
Naive Rebasement	36.8	-2.32	0.18	-2.14
No Regionalization	39.2	-1.31	0.06	-1.37
Conditional Regionalization	40.7	-2.96	0.08	-2.88
<i>Panel B: Mandatory MSSP</i>				
Simulated MSSP	100	-1.48	0.17	-1.31
No Rebasement	100	-2.35	0.19	-2.16
Naive Rebasement	100	-2.72	0.22	-2.50
No Regionalization	100	-1.78	0.07	-1.71
Conditional Regionalization	100	-3.55	0.11	-3.44

Notes: This table shows the percentage change in spending relative to the counterfactual scenario where the MSSP does not exist and all providers are under FFS (mandatory FFS). I weight the ACO-level spending data by the number of assigned beneficiaries, so that the statistics are representative of the average value across ACOs. Panel A shows the results under the voluntary MSSP participation, while Panel B shows the results under the mandatory MSSP participation. In both Panel A and B, columns one to four shows the following variables: (1) P represents the yearly average number of active ACOs from 2013 to 2022, (2) ΔY^I is the average percentage change in spending per-capita among of the ACOs that select in the MSSP excluding the Shared Savings Payments, (3) ΔY^O is the average percentage change in spending per-capita among of the ACOs that select out, (4) ΔY is the percentage change in spending per-capita among the all ACOs (5) ΔSS is the percentage change in Shared Savings Payments, and (6) ΔY is the percentage change in spending per-capita net of Shared Savings Payments among of the ACOs. In both Panel A and B, rows 1 report the results under the simulated MSSP regime with both benchmark rebasement and regionalization, and rows 2 to 5 report the results under the counterfactual four scenarios: (i) “No Rebasement” is the MSSP without the rebasement rule, (ii) in “Naive Rebasement” I consider MSSP with Benchmark Rebasement, but I impose that ACOs are not forward looking and maximize the static utility, (iii) “No Regionalization” is the MSSP without Benchmark Regionalization, and (iv) under “Conditional Regionalization”, in order to qualify for a positive regional adjustment, ACOs need to have generated savings in the previous Agreement Period.